

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Guzman Emmanuel

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

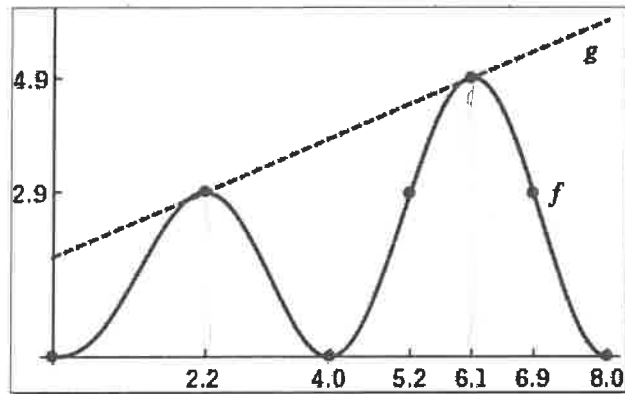
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = f(2.9)$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 0.0, 4.0, 8.0$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = (2.2, 2.9), (6.1, 4.9)$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$f(2.9) \quad 4.0 \leq \boxed{2.9} \leq 6.9$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V(x)$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$AV_{[a,b]} = \frac{V(b) - V(a)}{b - a}$$

$$AV_{[17,5]} = \frac{V(5) - V(17)}{5 - 17}$$

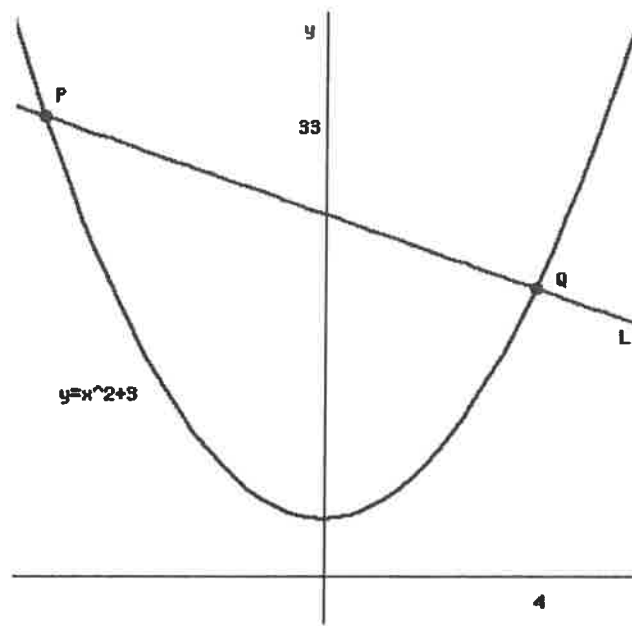
$$= -12$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

$$f(x)$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$P =$

$Q =$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

- (b) (2 pts) Determine the y -intercept of the parabola.

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

(b) (3 pts) $(g \circ f)(3)$

(c) (3 pts) $(f \circ f)(3)$

(d) (3 pts) $(g \circ g)(3)$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

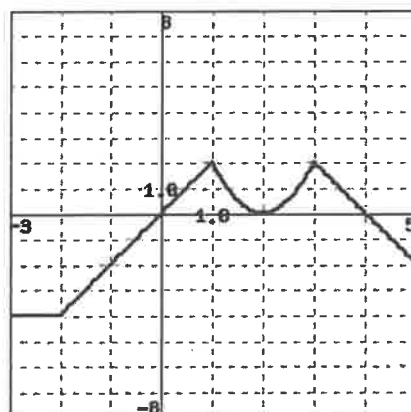
(b) (2 pts) Find $g(g^{-1}(2))$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2)$

(b) (3 pts) $w(2.5)$

(c) (3 pts) $k(1.5)$

(d) (3 pts) $n(-2.3)$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point _____

(b) (2 pts) The graph of $f(x) + 3$ must contain the point _____

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point _____

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

(b) (4 pts) $(f - g)(2)$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Killen, Sakaa W43280059

Please read the following instructions carefully.

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- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

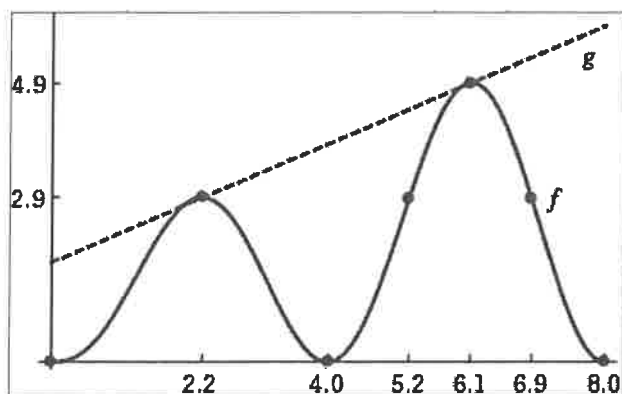
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 4.0, 8.0$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = (2.2, 2.9), (6.1, 4.9)$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$\begin{aligned} \text{AV}[a, b] &= \frac{f(b) - f(a)}{b - a} \\ a &= 4.0 \\ b &= 6.9 \\ &= \frac{f(6.9) - f(4.0)}{6.9 - 4.0} \\ &= 1 \end{aligned}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V =$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$V =$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

$$f(1) = 36 - 1^2$$

$$f(1) = 36 - 1$$

$$f(1) = 35$$

$$f(5) = 36 - 5^2$$

$$= 36 - 25$$

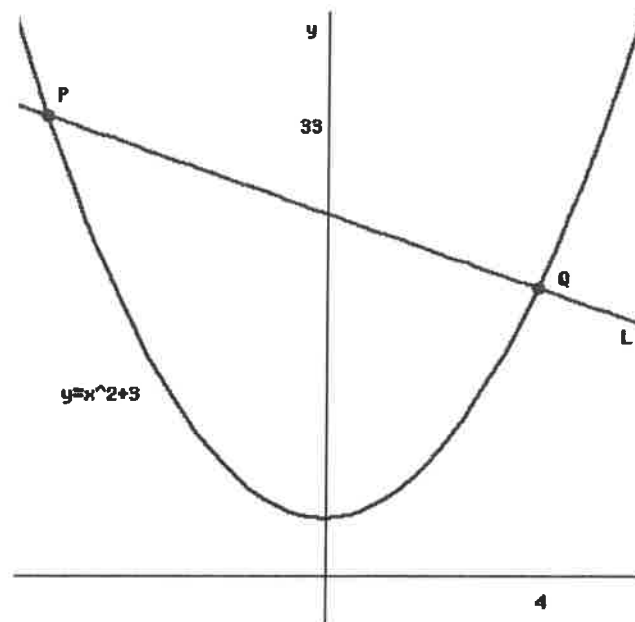
$$= 11$$

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

$$= \frac{35 - 11}{5 - 1}$$

$$= 3.8$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = \quad , 33$$

$$Q = 4, \quad$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$y = y_0 + m(x - x_0)$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$L =$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$y = a(x-h)^2 + k \quad \text{vertex } (h, k)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$(f \circ g) = 2x - 7 = 2(x + 5) - 7 = 2(3 + 5) - 7 = 2(8) - 7 =$$

(b) (3 pts) $(g \circ f)(3)$

$$g \circ f = (2x - 7) + 5 = (2(3) - 7) + 5 = (6 - 7) + 5 = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f \circ f = 2(2x - 7) - 7 = 2(2(3) - 7) - 7 = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$g \circ g = (x + 5) + 5 = (3 + 5) + 5 = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) =$$

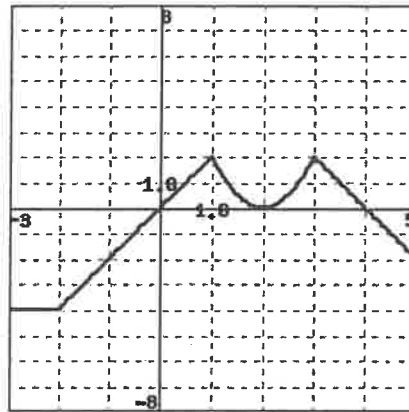
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9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = (30x + 597)^{-1}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1) = 1$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5 = -5$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) = 3$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = -3$$

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$$f(-5-2) = -7$$

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$$f(-5) + 3 = -2$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point _____

$$f(-5+5) - 4 = -4$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$((x^2 - 5) \cdot (4x + 1)) = 3$$

(b) (4 pts) $(f - g)(2)$

$$((x^2 - 5) - (4x + 1)) = 2$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\left(\frac{x^2 - 5}{4x + 1}\right) = 0$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Jayden Tam WA3013088

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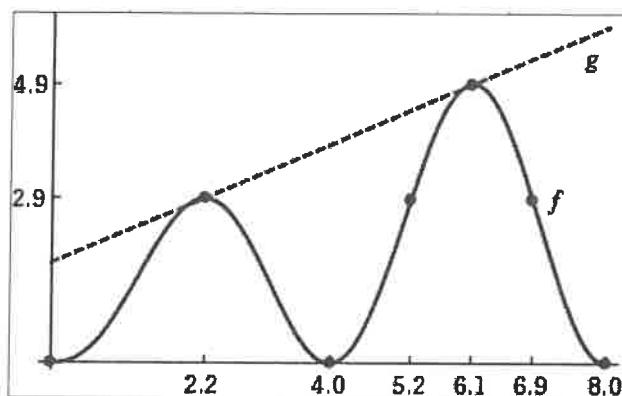
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

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- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4.0, 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.9, 2.2, 4.9, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$Av[4, 6.9] = \frac{f(6.9) - f(4.0)}{6.9 - 4.0} = \frac{2.9 - 0}{2.9} = 1$$

$4 = 0$

$6.9 = 2.9$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

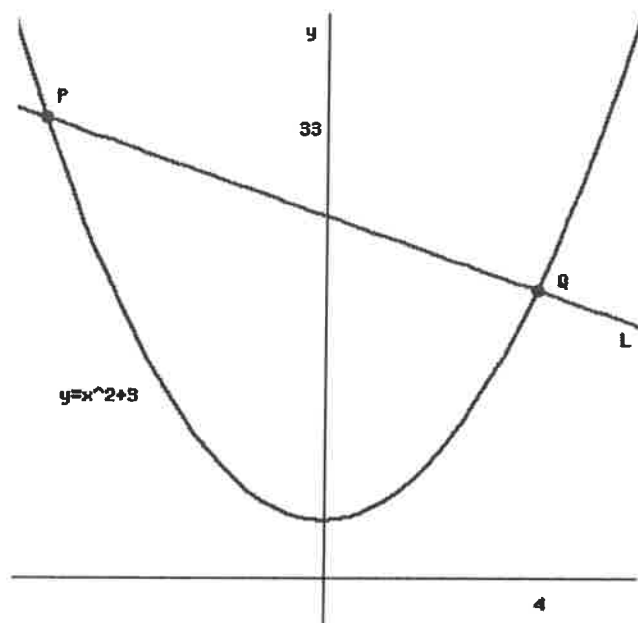
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$Av_{[1,5]} = \frac{f(1) - f(5)}{5 - 1} = \frac{-24}{4} = -6.$$

$$f(1) = 36 - (1)^2 = 35$$

$$f(5) = 36 - (5)^2 = 11$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



$$33 = C$$

- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-5.5, 33)$$

$$Q = (4, 19)$$

- (b) (4 pts) Use P and Q to determine the slope of the line L . $(-5.5, 33)$ $(4, 19)$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{19 - 33}{-5.5 - 4} = \frac{-14}{-9.5}$$

$$m = 1.4736$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$a=1 \quad b=5 \quad c=-14$$

$$\text{Using } x = -\frac{b}{2a}, \quad x = -\frac{5}{2(1)} = -2.5$$

$$\begin{aligned} f(-2.5) &= (-2.5)^2 + 5(-2.5) - 14 \\ &= 6.25 + -12.5 - 14 \\ &= -20.25 \end{aligned}$$

$$\text{So, } (-2.5, -20.25)$$

- (b) (2 pts) Determine the y-intercept of the parabola.

$$\text{Set } x=0$$

$$f(0) = (0)^2 + 5(0) - 14$$

$$(0)^2 = 0$$

$$5(0) = 0$$

$$0 - 0 - 14 = -14$$

$$\text{So, } (0, -14) \text{ is the y-intercept}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x-5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 3(x-5)^2 - 9$$

$$0 = 3(x-5)^2 - 9$$

$$0 + 9 = 3(x-5)^2 - 9 + 9$$

$$9 = 3(x-5)^2$$

$$\frac{9}{3} = \frac{3(x-5)^2}{3}$$

$$3 = (x-5)^2$$

$$\pm \sqrt{3} = x - 5$$

$$x = -5 \pm \sqrt{3}$$

$$x = -5 + \sqrt{3}, \quad x = -5 - \sqrt{3}$$

$$\text{Set } y=0 \text{ and solve for } x$$

$$\text{Add } 9 \text{ to both sides}$$

$$-9 + 9 = 0 \text{ so they cancel out}$$

$$\text{Divide each side by } 3$$

$$\text{Take the square root of both sides}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3) = f(g(3))$ Find $g(3)$

$$g(3) = (3) + 5 = 8$$

$$f(8) = 2(8) - 7 = 9$$

(b) (3 pts) $(g \circ f)(3) = g(f(3))$

Find $f(3) = 2(3) - 7$
 $= -1$

$$g(-1) = (-1) + 5 = 4$$

(c) (3 pts) $(f \circ f)(3) = f(f(3))$

$$f(3) = 2(3) - 7$$

$$= -1$$

$$f(-1) = 2(-1) - 7$$

$$= -9$$

(d) (3 pts) $(g \circ g)(3)$

Find $g(3)$

$$g(3) = (3) + 5 = 8$$

$$g(8) = (8) + 5 = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0) = 1$

Looking at the row for $g(x)$ in the table, we see that $g(x) = 0$ when $x = 1$

(b) (2 pts) Find $g(g^{-1}(2))$ $g^{-1}(2) = 2$

Then substitute that back into $g \rightarrow g(2) = 2$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

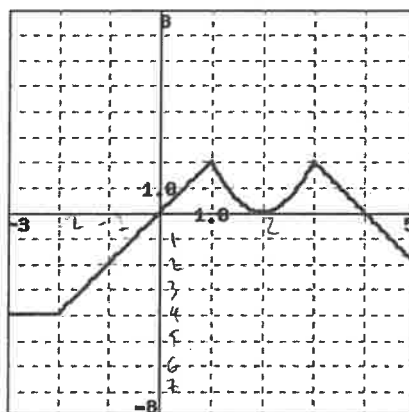
$$g^{-1}(C) = C = 30x + 597$$

$$C - 597 = 30x$$

$$x = \frac{C - 597}{30}$$

$$g^{-1}(C) = \frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



$$m = -4, 0, 1, 1$$

$$t = -3, 0, 1, 5$$

Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1), \quad m(2 - 1) = 0$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5)$$

$$= m(2.5 - 0.5)$$

$$= 2$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5)$$

$$= m(1.5 + 1.5)$$

$$= 1.5$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2$$

$$= m(-2.3) + 2$$

$$= -0.3$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x-2)$ must contain the point $(-3, 7)$

Transformation - Shifts 2 units to the right

Translation - Add 2 to the x-coordinate $(-5+2, 7)$

$$-5+2 = -3$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

Transformation - Vertical Shifts 3 units up

Translation $(-5, 7+3)$ $7+3=10$

(c) (2 pts) The graph of $f(x+5) - 4$ must contain the point _____

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) = f(x) \cdot g(x)$

$$\begin{array}{rcl} (-3)^2 - 5 & \cdot & 4(-3) + 1 \\ 4 & & -11 \\ \hline & & -44 \end{array}$$

(b) (4 pts) $(f - g)(2) = f(x) - g(x)$

$$\begin{array}{rcl} (2)^2 - 5 & - & 4(2) + 1 \\ -1 & & 17 \\ \hline & & -18 \end{array}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0) = \frac{f(x)}{g(x)}$

$$\begin{array}{rcl} (0)^2 - 5 & & \\ \hline 4(0) + 1 & & \\ \hline & & -5 \end{array}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Giovanni Galcomazzi WA3268356

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

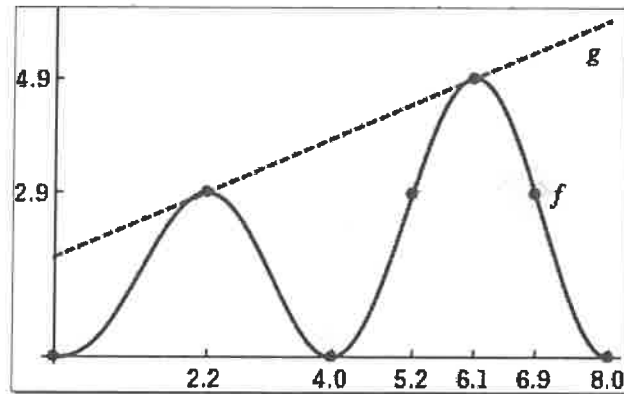
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{y-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 4.0, x = 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, x = 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

$a = 4.0$ $b = 6.9$

$f(4.0) = 0 = a$

$f(6.9) = 2.9 = b$

$$AV = \frac{2.9 - 0}{6.9 - 4.0}$$

$= \frac{2.9}{2.9}$

$= \frac{2.9}{2.9}$

$= 1$

$AV = 1$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = L \cdot W \cdot H$$

$$v(x)$$

$$17 - 2x = L$$

$$5 - 2x = W$$

$$+1 = x$$

$$v(x) = (17 - 2x)(5 - 2x)(x)$$



- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$5 - 2x > 0 \rightarrow x < 2.5$$

$$17 - 2x > 0 \rightarrow x < 8.5$$

$$x > 0$$

$$D = (0, 2.5)$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

$$a = 1_{x_1} \quad b = 5_{x_2}$$

$$= \frac{f(5) - f(1)}{5 - 1}$$

$$f(x) = 36 - x^2$$

$$f(1) = 36 - 1^2$$

$$f(5) = 36 - 5^2$$

$$= \frac{(36 - 5^2) - (36 - 1^2)}{5 - 1}$$

$$(36 + 10) - (36 + 1)$$

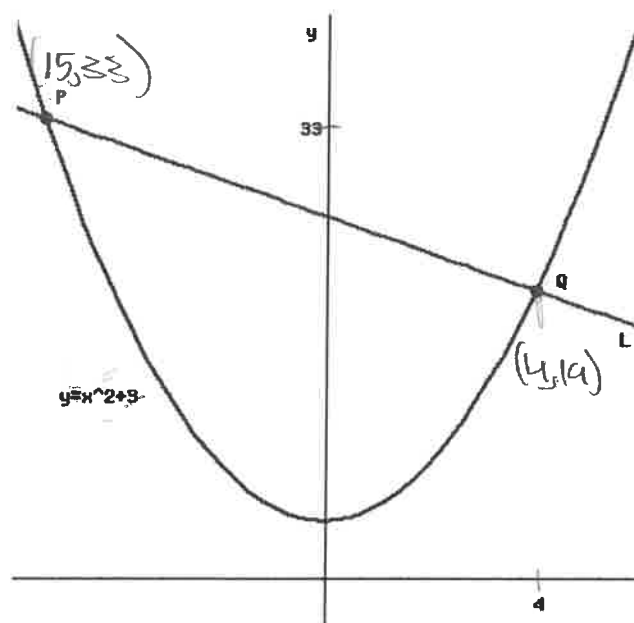
$$5 - 1$$

$$(46) - (37)$$

$$4$$

$$AV = \frac{9}{4} = 2.25$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$P = (15, 33)$
 $Q = (4, 19)$
 $y = x^2 + 3$
 $33 = x^2 + 3$
 -3
 $30 = x^2$
 $\frac{30}{2} = \frac{x^2}{2}$
 $15 = x$
 $y = 4^2 + 3$
 $y = 16 + 3$
 $y = 19$

Point Slope form
 $y = y_0 + m(x - x_0)$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$y_1 = y_2 + m(x_1 - x_2)$
 $33 = 19 + m(15 - 4)$
 $33 = 19 + m(11)$
 $33 = 19 + m(11)$
 $33 = 19 + 11m$
 $14 = 11m$
 $m = \frac{14}{11}$

$33 = 19 + m(11)$
 $33 = 19 + 11m$
 $14 = 11m$
 $m = \frac{14}{11}$

$33 = 19 + m(11)$
 $33 = 19 + 11m$
 $14 = 11m$
 $m = \frac{14}{11}$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$33 =$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

$\begin{matrix} a & b & c \end{matrix}$

- (a) (6 pts) Determine the vertex of the parabola.

$$\text{Vertex } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

$$\begin{aligned} & \left(-\frac{5x}{2x^2}, q\left(-\frac{5x}{2x^2}\right) \right) \\ & \left(-2x^2, q(-2x^2) \right) \\ & x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5x \pm \sqrt{x^2 - 14}}{2x^2} \\ & x = \frac{-5x \pm \sqrt{5x^2 - 4x^2 - 14}}{2x^2} \end{aligned}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$(f \circ g)(3) = f(g(3))$$

$$f(8)$$

$$f(3) = 2(3) - 7$$

$$f(3) = 2(3) - 7$$

$$= 6 - 7$$

$$f(3) = 1$$

$$g(3) = 3 + 5$$

$$f(8) = 2(8) - 7$$

$$f(8) = 16 - 7$$

$$= 9$$

$$g(3) = 3 + 5$$

$$g(3) = 8$$

(b) (3 pts) $(g \circ f)(3)$

$$g(f(3))$$

$$g(1) = 1 + 5$$

$$g(1) = 1 + 5$$

$$= 6$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3))$$

$$f(1) = 2(1) - 7$$

$$f(1) = 2(1) - 7$$

$$2 - 7$$

$$= -5$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3))$$

$$g(8) = 8 + 5$$

$$g(8) = 13$$

$$g(8) = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

$$3^{-1} = -9$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2))$$

$$g^{-1}(2) = 2$$

$$2^{-1} = -4$$

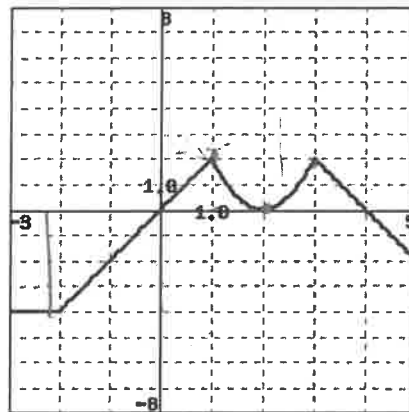
$$g(2) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$\cancel{P(2)} \quad P(2) = m(2-1) \\ P(2) = m(1) = \boxed{2}$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5)$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5)$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2$$

$$m(-2.3) = -4 + 2$$

$$n(-2.3) = m(-2.3) + 2 = \boxed{-2}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x-2)$ must contain the point $(-7, 7)$
horizontal translation $(-5-2, 7)$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$
vertical translation $(x, y+a)$

(c) (2 pts) The graph of $f(x+5) - 4$ must contain the point $(0, 3)$
 $(x+a, y+b)$
 $(-5+5, 7-4) = (0, 3)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) \cdot g(-3) =$$

$$1 \cdot -11 = \boxed{-11}$$

$$f(-3) = -3^2 - 5$$

$$= 6 - 5$$

$$f(-3) = 1$$

$$g(-3) = 4(-3) + 1$$

$$= -12 + 1$$

$$g(-3) = -11$$

(b) (4 pts) $(f - g)(2)$

$$f(2) - g(2) =$$

$$-1 - 9 = \boxed{-10}$$

$$f(2) = 2^2 - 5$$

$$= 4 - 5$$

$$f(2) = -1$$

$$g(2) = 4(2) + 1$$

$$= 8 + 1$$

$$g(2) = 9$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$f(0) = 0^2 - 5$$

$$= -5$$

$$\left(\frac{f}{g}\right)$$

$$\frac{-5}{1} = \boxed{-5}$$

$$g(0) = 4(0) + 1$$

$$= 1$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Gonzalez, Cristhian, WA3009254

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$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

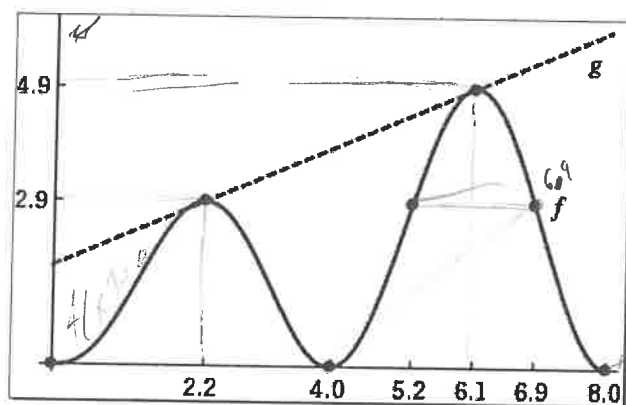
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
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1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = g(5.2)$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9)

(6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = (4.0, 0), (8.0, 0)$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

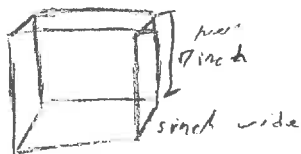
$x = (2.2, 2.9), (6.1, 4.9)$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

The average rate of $f(x) = 1$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.



length ~

$$L = 17$$

$$W = 5$$

height

x
volume

$$V = x(L - 2x)(W - 2x)$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$\text{height} > 0$$

$$\text{length} = 17 - 2x > 0 \Rightarrow 2x < 17 \Rightarrow x < 8.5$$

$$6 - 2x > 0 \Rightarrow 2x < 6 \Rightarrow x < 3$$

$$(0, 3)$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

$$f(1) = 36 - 1^2 = 36 - 1 = 35$$

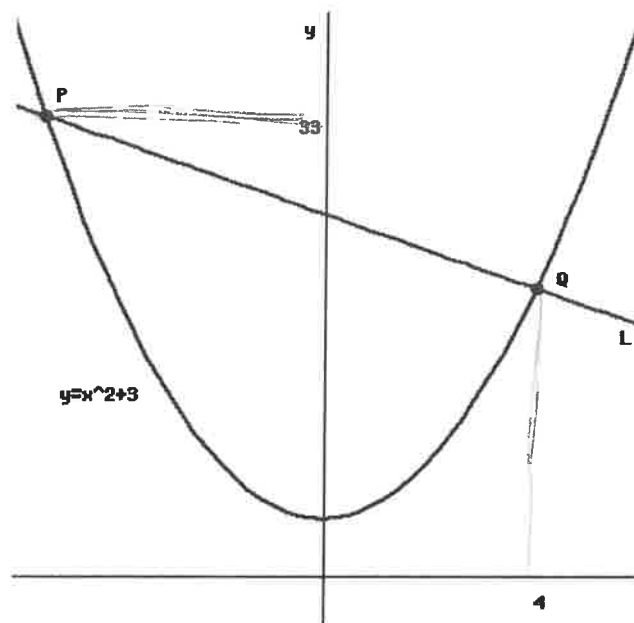
$$f(5) = 36 - 5^2 = 36 - 25 = 11$$

$$\begin{array}{r} 36 \\ - 25 \\ \hline 11 \end{array}$$

$$\begin{array}{r} 11 - 35 \\ \hline -24 \\ 5 - 1 \\ \hline 4 \end{array}$$

$$\frac{-24}{4} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = y = (4)^2 + 3 = 16 + 3 = 19$$

$$Q = (4, 19)$$

$$y = x^2 + 3$$

$$33 = x^2 + 3$$

$$x^2 = 30$$

$$x = \pm\sqrt{30}$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{19 - 33}{4 - (-\sqrt{30})} = \frac{-14}{4 + \sqrt{30}}$$

$$\frac{-14(4 - \sqrt{30})}{-14(56 - \sqrt{30})}$$

$$\frac{-56 + 14\sqrt{30}}{16 - 30} = \frac{-56 + 14\sqrt{30}}{-14}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = x^2 + 3$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$f(x) = x^2 + 5x - 14$$

$$h = -\frac{b}{2a}$$

$$k = f\left(-\frac{5}{2}\right) = \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14$$

$$a = 1 \quad h = -\frac{5}{2(1)} = -\frac{5}{2} = -2.5 \quad k = \frac{25}{4} - \frac{25}{2} - 14$$

$$b = 5$$

$$k = -\frac{81}{4} = -20.25$$

$$k = \frac{25}{4} - \frac{50}{4} - \frac{56}{4}$$

- (b) (2 pts) Determine the y-intercept of the parabola.

$$f(x) = x^2 + 5x - 14$$

$$f(0) = 0^2 + 5(0) - 14$$

$$0 = 0 + 0 - 14$$

$$f = \boxed{0, -14}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 3(x - 5)^2 - 9$$

$$3(x - 5)^2 - 9 = 0$$

$$3(x - 5)^2 = \frac{9}{3}$$

$$(x - 5)^2 = \frac{9}{3}$$

$$x - 5 = \pm \sqrt{\frac{9}{3}}$$

$$\boxed{\sqrt{3-5}}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

(b) (3 pts) $(g \circ f)(3)$

(c) (3 pts) $(f \circ f)(3)$

(d) (3 pts) $(g \circ g)(3)$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2)) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

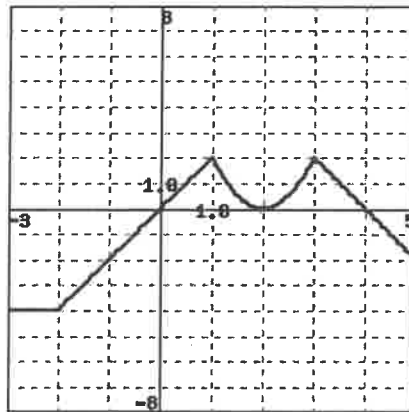
$$f(g(x)) = 6$$

$$x = 0, 4$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1)$$

$$p(2) = m(1)$$

$$p = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5$$

$$= m(2) - 2.5$$

$$(2) - 2.5 = -0.5$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) = m(3)$$

$$= 4.5$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = 2 = 4.6$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $-7, 7$

$$f(-5-2) = -7$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $-2, 7$

$$f(-5) + 3 = -2$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $-4, 7$

$$f(-5+5) - 4 = -4$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

(b) (4 pts) $(f - g)(2)$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Fang, Swyi NA3005435

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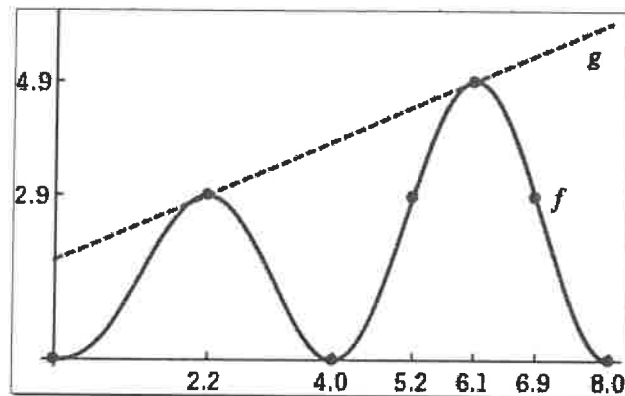
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

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1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = \boxed{2.9}$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, \boxed{2.9}) \quad (\boxed{6.1}, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = \boxed{0, 4, 8}$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = \boxed{2.2, 6.1}$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$AV_{[4.0, 6.9]} = \frac{2.9 - 0}{6.9 - 4.0} = \frac{2.9}{2.9} = \boxed{1}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = LWH$$

$$V(x) = x(17 - 2x)(5 - 2x)$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$H = x > 0$$

$$W = 5 - 2x > 0$$

$$5 > 2x$$

$$\frac{5}{2} > x$$

$$0 < x < \frac{5}{2}$$

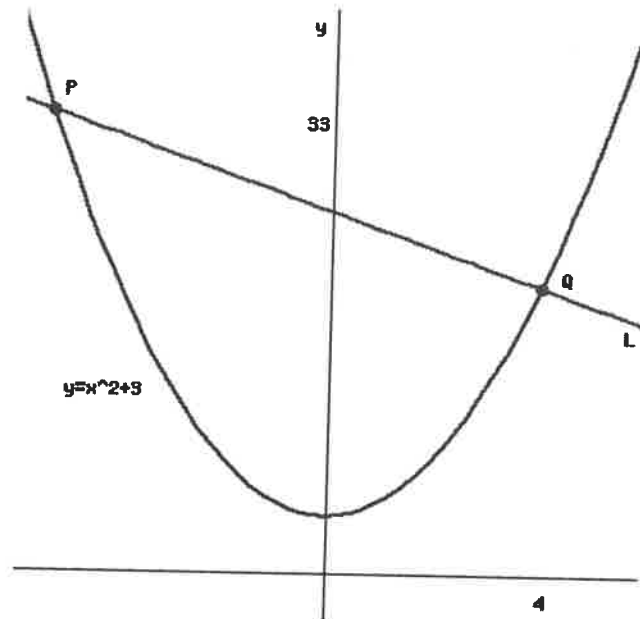
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$\begin{aligned} f(1) &= 36 - 1^2 \\ &= 35 \end{aligned}$$

$$\begin{aligned} f(5) &= 36 - 5^2 \\ &= 36 - 25 \\ &= 11 \end{aligned}$$

$$\begin{aligned} AV_{[1, 5]} &= \frac{11 - 35}{5 - 1} = \frac{-24}{4} \\ &= \boxed{-6} \end{aligned}$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

$$y = 4^2 + 3$$

$$= 19$$

$$33 = x^2 + 3$$

$$\sqrt{30} = \sqrt{x^2}$$

$$\sqrt{30} \pm = x$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = \frac{19 - 33}{4 - (-\sqrt{30})} = \frac{-14}{4 + \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y - y_1 = m(x - x_1)$$

$$L = 33 - \frac{14}{4 + \sqrt{30}} (x + \sqrt{30})$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

$\begin{matrix} a & b & c \end{matrix}$

- (a) (6 pts) Determine the vertex of the parabola.

$$\begin{aligned} x &= \frac{-b}{2a} \\ &= \frac{-5}{2(1)} \\ &= -\frac{5}{2} \end{aligned}$$

$$\begin{aligned} y &= \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14 \\ &= \frac{25}{4} - \frac{25}{2} - 14 \\ &= \frac{25}{4} - \frac{50}{4} - \frac{56}{4} \\ &= \frac{-81}{4} \end{aligned}$$

$$\boxed{\left(-\frac{5}{2}, -\frac{81}{4}\right)} \\ \boxed{(-2.5, -20.25)}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$\begin{aligned} f(0) &= 0^2 + 5(0) - 14 \\ &= 0 + 0 - 14 \\ &= -14 \end{aligned}$$

$$\boxed{(0, -14)}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\begin{aligned} 0 &= 3(x - 5)^2 - 9 \\ 9 &= 3(x - 5)^2 \\ 3 &= (x - 5)^2 \\ \pm \sqrt{3} &= x - 5 \end{aligned}$$

$$\begin{aligned} x - 5 &= \sqrt{3} & x - 5 &= -\sqrt{3} \\ \boxed{x = \sqrt{3} + 5} & & \boxed{x = -\sqrt{3} + 5} \end{aligned}$$

$$\begin{aligned} &3(x - 5)^2 - 9 \\ &(3x - 15)^2 - 9 \\ &(3x - 15)(3x - 15) - 9 \\ &9x^2 - 45x - 45x + 225 - 9 \\ &9x^2 - 90x + 216 \end{aligned}$$

$$\begin{aligned} x &= \frac{-(-90) \pm \sqrt{(-90)^2 - 4(9)(216)}}{2(9)} \\ &= \frac{90 \pm \sqrt{324}}{18} \end{aligned}$$

$$\begin{aligned} x &= \frac{90 + \sqrt{324}}{18} & x &= \frac{90 - \sqrt{324}}{18} \\ &= 6 & &= 4 \end{aligned}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$ $f(g(3)) = \boxed{9}$

$$\begin{aligned} g(3) &= 3 + 5 \\ &= 8 \end{aligned} \quad \begin{aligned} f(8) &= 2(8) - 7 \\ &= 16 - 7 \\ &= 9 \end{aligned}$$

(b) (3 pts) $(g \circ f)(3)$

$$\begin{aligned} g(f(3)) &= \boxed{4} \\ f(3) &= 2(3) - 7 \\ &= 6 - 7 \\ &= -1 \end{aligned}$$

$$\begin{aligned} g(-1) &= -1 + 5 \\ &= 4 \end{aligned}$$

(c) (3 pts) $(f \circ f)(3)$

$$\begin{aligned} f(f(3)) &= \boxed{-9} \\ f(-1) &= 2(-1) - 7 \\ &= -2 - 7 = -9 \end{aligned}$$

(d) (3 pts) $(g \circ g)(3)$

$$\begin{aligned} g(g(3)) &= \boxed{13} \\ g(8) &= 8 + 5 \\ &= 13 \end{aligned}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0) = \boxed{1}$

(b) (2 pts) Find $g(g^{-1}(2)) = \boxed{2}$

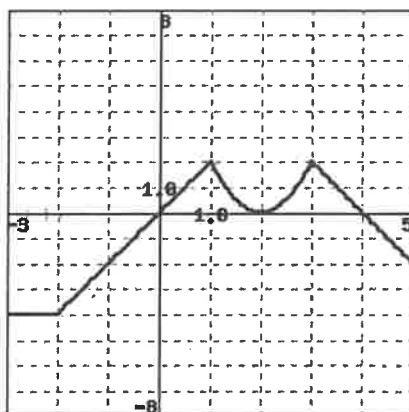
(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$? $\boxed{0, 4}$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{C - 597}{30}$$

$$\begin{aligned} C &= 30x + 597 \\ C - 597 &= 30x \\ \frac{C - 597}{30} &= x \end{aligned}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2) = \boxed{2}$

$$\begin{aligned} p(2) &= m(2 - 1) \\ &= m(1) \end{aligned}$$

(b) (3 pts) $w(2.5) = \boxed{-2.5}$

$$\begin{aligned} w(2.5) &= m(2.5 - 0.5) \\ &= m(2) - 2.5 \\ &= 0 - 2.5 \\ &= -2.5 \end{aligned}$$

(c) (3 pts) $k(1.5) = \boxed{2}$

$$\begin{aligned} k(1.5) &= m(1.5 + 1) \\ &= m(2.5) \\ &= 2 \end{aligned}$$

(d) (3 pts) $n(-2.3) = \boxed{-2}$

$$\begin{aligned} n(-2.3) &= m(-2.3) + 2 \\ &= -4 + 2 \\ &= -2 \end{aligned}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x-2)$ must contain the point $(1, 7)$

$$-5 + 6 = 1$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

$$7 + 3 = 10$$

(c) (2 pts) The graph of $f(x+5) - 4$ must contain the point $(-10, 3)$

$$-5 - 5 = -10$$

$$7 - 4 = 3$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) = -44$

$$(4) \cdot (-11) = -44$$

(b) (4 pts) $(f - g)(2) = -10$

$$-1 - 9 = -10$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0) = \frac{-5}{1} = -5$

$$\begin{aligned} f(-3) &= -3^2 - 5 \\ &= 9 - 5 \\ &= 4 \end{aligned}$$

$$\begin{aligned} g(-3) &= 4(-3) + 1 \\ &= -12 + 1 \\ &= -11 \end{aligned}$$

$$\begin{aligned} f(2) &= 2^2 - 5 \\ &= 4 - 5 \\ &= -1 \end{aligned}$$

$$\begin{aligned} g(2) &= 4(2) + 1 \\ &= 8 + 1 \\ &= 9 \end{aligned}$$

$$\begin{aligned} f(0) &= 0^2 - 5 \\ &= -5 \end{aligned}$$

$$\begin{aligned} g(0) &= 4(0) + 1 \\ &= 1 \end{aligned}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Burnoski, Elsiebeth WA 3279774

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

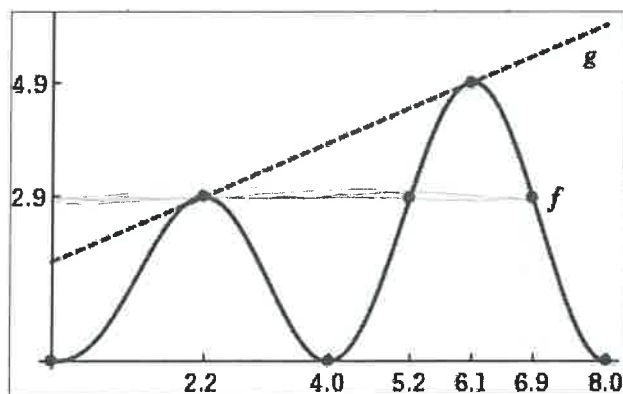
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4.0, 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

$$\frac{f(6.9) - f(4.0)}{6.9 - 4.0}$$

$$f(6.9) = 2.9$$

$$f(4.0) = 0$$

$$\frac{2.9 - 0}{6.9 - 4.0} = \frac{2.9}{2.9} = 1$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

length $\times$ width
17 inches $\times$ 5 inches

volume = length $\cdot$ width $\cdot$ height

$$17 - x - x = 17 - 2x$$

$$5 - x - x = 5 - 2x$$

$$V(x) = (17 - 2x)(5 - 2x)(x)$$

$$V(x) = x(17 - 2x)(5 - 2x)$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

means: what values x can actually have when we are physically making the box.

$$x > 0$$

$$5 - 2x$$

$$5 - 2x > 0$$

$$5 > 2x$$

$$2.5 > x$$

$$x < 2.5$$

$$(0, 2.5)$$

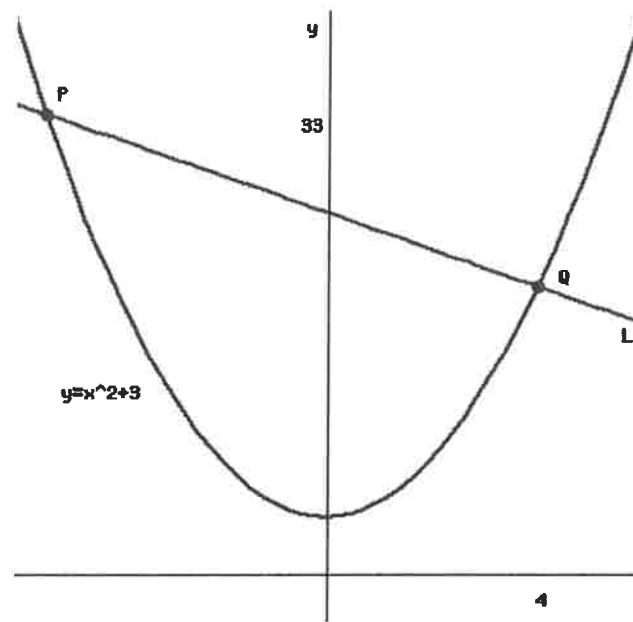
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$$[1, 5] = f(1) = 36 - 1^2 = 35$$

$$f(5) = 36 - 5^2 = 11$$

$$\frac{11 - 35}{5 - 1} = -\frac{24}{4} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



point P

$$\begin{aligned} 33 &= x^2 + 3 \\ 30 &= x^2 \\ x &= \pm \sqrt{30} \\ \text{so} \\ (-\sqrt{30}, 33) \end{aligned}$$

- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

point Q

$$\begin{aligned} y &= x^2 + 3 \\ y &= (4)^2 + 3 \\ y &= 16 + 3 \\ y &= 19 \\ \text{so } (4, 19) \end{aligned}$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = \frac{y^2 - y^1}{x^2 - x^1}$$

$$m = \frac{19 - 33}{4 - (-\sqrt{30})}$$

$$m = \frac{-14}{4 + \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y - y_1 = m(x - x_1)$$

$$(4, 19)$$

$$y - 19 = \frac{-14}{4 + \sqrt{30}}(x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$f(x) = ax^2 + bx + c$$

$$x = -\frac{b}{2a}$$

$$a = 1, b = 5, c = -14$$

$$x = \frac{-(5)}{2(1)} = -\frac{5}{2}$$

$$\text{vertex} = \left(-\frac{5}{2}, -\frac{81}{4}\right)$$

$$f\left(-\frac{5}{2}\right) = \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14$$

$$-\frac{5}{1} \times -\frac{5}{2} = \frac{25}{2}$$

$$\frac{25}{4} - \frac{50}{4} - \frac{56}{4} = \frac{81}{4} = +\frac{25}{4} + \frac{25}{2} - 14$$

$$\frac{14}{1} \times \frac{4}{4} = \frac{56}{4}$$

$$= +\frac{25}{4} - \frac{50}{4} - \frac{56}{4} = -\frac{81}{4}$$

- (b) (2 pts) Determine the y-intercept of the parabola.

$$x = 0$$

$$f(0) = 0^2 + 5(0) - 14$$

$$f(0) = -14$$

$$\text{y-intercept} = (0, -14)$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 0$$

$$0 = 3(x - 5)^2 - 9$$

$$9 = 3(x - 5)^2$$

$$x + 5 = \pm \sqrt{3}$$

$$\frac{-b}{2(a)}$$

$$\begin{aligned} x &= -5 + \sqrt{3} \\ &\$ \\ x &= -5 - \sqrt{3} \end{aligned}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3) = f(g(3))$
 $g(3) = 3 + 5 = 8$
 $f(8) = 2(8) - 7 = 16 - 7 = \boxed{9}$

(b) (3 pts) $(g \circ f)(3) = g(f(3))$
 $f(3) = 2(3) - 7$
 $6 - 7 = -1$
 $g(-1) = -1 + 5 = 4 = \boxed{4}$

(c) (3 pts) $(f \circ f)(3) = f(f(3))$
 $f(3) = -1$
 $f(-1) = 2(-1) - 7 = -2 - 7 = \boxed{-9}$

(d) (3 pts) $(g \circ g)(3) = g(g(3))$
 $g(3) = 8$
 $g(8) = 8 + 5 = \boxed{13}$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0) = \boxed{1}$

$X = 1 ; (3, 0, 2, 4, 1)$

(b) (2 pts) Find $g(g^{-1}(2)) = \boxed{2}$

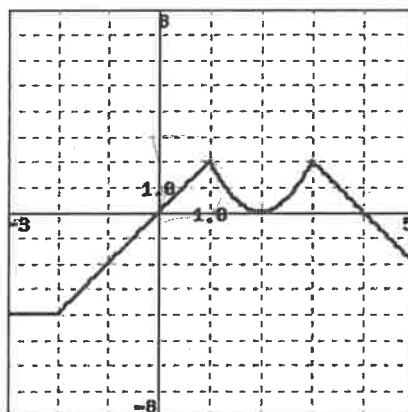
(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$? $= \boxed{0, 4}$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \left\lfloor \frac{C - 597}{30} \right\rfloor$$

$$\frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2) = \boxed{2}$

$$\begin{aligned} p(t) &= m(t - 1) \\ p(2) &= m(2 - 1) \\ p(2) &= m(1) \end{aligned}$$

(b) (3 pts) $w(2.5) = \boxed{-2.5}$

$$\begin{aligned} w(t) &= m(t - 0.5) \\ w(2.5) &= m(2.5 - 0.5) - 2.5 \\ w(2.5) &= m(2) - 2.5 \end{aligned}$$

(c) (3 pts) $k(1.5) = \boxed{2}$

$$\begin{aligned} k(t) &= m(t + 1.5) \\ k(1.5) &= m(1.5 + 1.5) \end{aligned}$$

(d) (3 pts) $n(-2.3) = \boxed{-2}$

$$k(1.5) = 3$$

$$n(t) = m(t) + 2$$

$$\begin{aligned} n(-2.3) &= m(-2.3) + 2 \\ &= m(-0.3) \end{aligned}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs? $f(-5) = 7$

(a) (2 pts) The graph of $f(x-2)$ must contain the point $(-3, 7)$

$$\begin{array}{c} x-2 \\ (-5, 7) \end{array} \quad \begin{array}{l} \text{add 2 to the} \\ x \text{ coordinate} \end{array}$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

(c) (2 pts) The graph of $f(x+5) - 4$ must contain the point $(-10, 3)$

$$\begin{array}{l} x+5 = \text{move 5} \\ -5-5 = -10 \end{array}$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) = \boxed{154}$ $f(-3)?$

$$\begin{array}{l} (f \cdot g)(-3) = f(-3) \cdot g(-3) \quad f(-3) = -3^2 - 5 \\ g(-3) = 4(-3) + 1 = -11 \cdot -14 \quad -9 - 5 = -14 \end{array}$$

(b) (4 pts) $(f - g)(2)$

$$\begin{array}{l} f(2) - g(2) \quad f(2) = 2^2 - 5 = 4 - 5 = -1 \quad -1 - 9 = \boxed{-10} \\ g(2) = 4(2) + 1 = 9 \quad 8 + 1 \end{array}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\begin{array}{l} \frac{f(0)}{g(0)} \quad f(0) = 0^2 - 5 = -5 \\ g(0) = 4(0) + 1 = 1 \\ \frac{-5}{1} = \boxed{-5} \end{array}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#:

Elijah Hankins III 560822646

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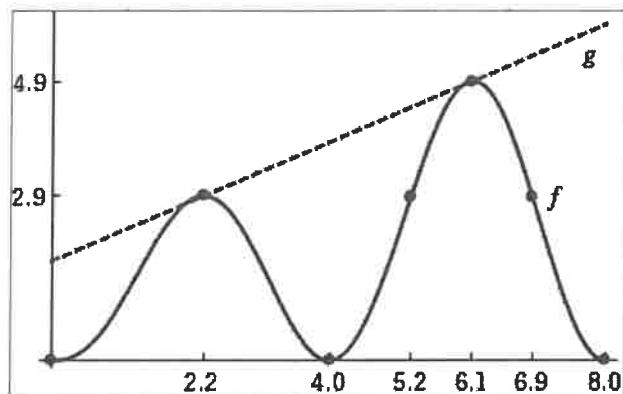
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1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = (4.0, 0), (8.0, 0)$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

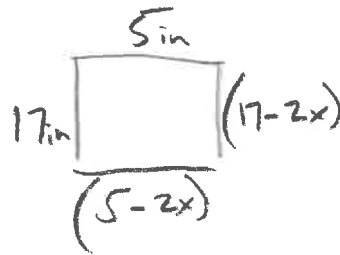
$x = (2.2, 6.1)$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

AV $\left[\frac{f(b) - f(a)}{b - a} \right] = \frac{f(6.9) - f(4.0)}{6.9 - 4.0} = \frac{2.9 - 0}{2.9} = \frac{2.9}{2.9} = 1$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.



$$V = \text{width} \cdot \text{length} \cdot \text{height}$$

$$V = (5-2x)(17-2x)(x)$$

$$V = x(5-2x)(17-2x)$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$\begin{aligned} x &> 0 \\ 5-2x &> 0 \\ -2x &> -5 \\ \frac{-2x}{-2} &> \frac{-5}{-2} \\ x &< 2.5 \end{aligned}$$

$$\begin{aligned} x &> 0 \\ 17-2x &> 0 \\ -2x &> -17 \\ \frac{-2x}{-2} &> \frac{-17}{-2} \\ x &< 8.5 \end{aligned}$$

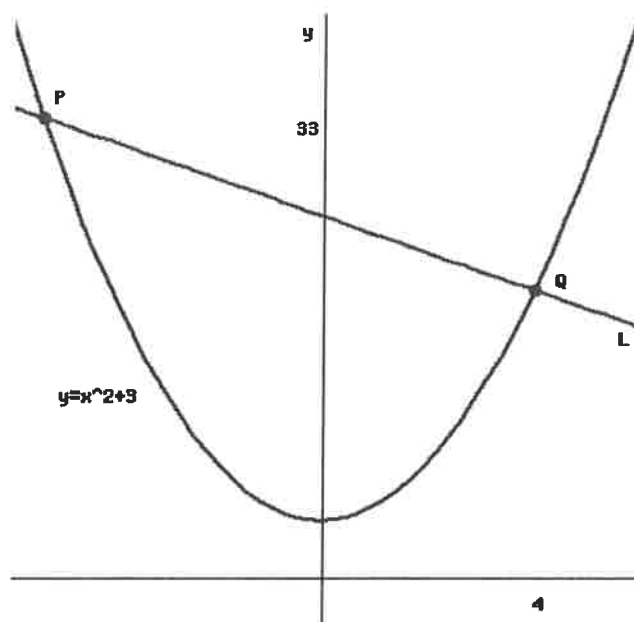
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$$\text{Avg}_{[1,5]} = \frac{f(5) - f(1)}{5 - 1} = \frac{11 - 35}{4} = \frac{-24}{4} = -6$$

$$f(1) = 36 - (1)^2 = 35$$

$$f(5) = 36 - (5)^2 = 11$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

$$y = (4)^2 + 3 = 19$$

$$Q(4, 19)$$

$$P = (-\sqrt{30}, 33)$$

$$33 = x^2 + 3$$

$$30 = x^2$$

$$\pm\sqrt{30} = \sqrt{x^2}$$

$$\pm\sqrt{30} = x$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$AV[-\sqrt{30}, 4] = \frac{f(4) - f(-\sqrt{30})}{4 - (-\sqrt{30})} = \frac{19 - 33}{4 + \sqrt{30}} = \frac{-14}{4 + \sqrt{30}}$$

$$m = AV = \frac{-14}{4 + \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = L = y_0 + m(x - x_0)$$

$$y = L = 19 - \frac{14}{4 + \sqrt{30}}(x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14. \quad (a=1, b=+5), (c=-14)$$

- (a) (6 pts) Determine the vertex of the parabola.

$$V = \left(\frac{-b}{2a}, y\left(\frac{-b}{2a}\right) \right)$$

$$\frac{5}{2(1)} = \frac{5}{2} = 2.5$$

$$g(2.5) = (2.5)^2 - 5(2.5) - 14$$

$$= -20.25$$

$$V = 2\frac{1}{2}, -20.25$$

- (b) (2 pts) Determine the y-intercept of the parabola.

x-intercepts.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5 \pm \sqrt{5^2 - 4(1)(-14)}}{2(1)}$$

$$x = \frac{5 + 5.5}{2} = 10.25 \quad \text{or} \quad x = \frac{5 - 5.5}{2} = -0.25$$

$(-0.25, 10.25)$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$3(x - 5)^2 - 9 = 0$$

$$3(x - 5)^2 = 9$$

$$(x - 5) = 3$$

$$\sqrt{(x - 5)^2} = \pm \sqrt{3}$$

$$x - 5 =$$

$$x = 2.5$$

$$= 10.25 = y\text{-intercept}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(3) = 5 - 6$$

$$f(3) = 15 - 6 = \boxed{9}$$

$$g(3) = 3 + 5$$

$$g(3) = (3) + 5$$

$$g(3) = \boxed{8}$$

(b) (3 pts) $(g \circ f)(3)$

$$\boxed{8}$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3))$$

$$f(9)$$

$$= 5(9) - 6 = \boxed{39}$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3))$$

$$g(8)$$

$$2(4) - 2 = \boxed{6}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$\boxed{1}$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2))$$

$$= g(0)$$

$$= \boxed{1}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(g(x)) = 6$$

$$f(6) = 6$$

$$f(4) = 6$$

$$\boxed{x = 2, 3}$$

$$g(2) = 6$$

$$g(3) = 4$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$g^{-1}(C) =$

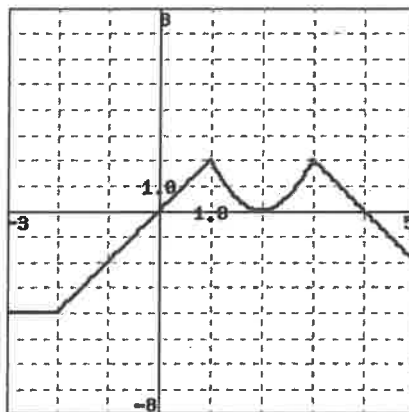
$$C = 597 + 30x$$

$$C - 597 = 30x$$

$$\frac{C - 597}{30} = \frac{30x}{30}$$

$$\frac{1}{30}C - 19.9 = x = g^{-1}(C) = \frac{1}{30}C - 19.9$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$\boxed{3}$$

$$p(2) = m(2-1) = m(1) = 2$$

- (b) (3 pts) $w(2.5)$

$$\boxed{-3}$$

$$\begin{aligned} w(2.5) &= m(2.5 - 0.5) - 2.5 \\ &= m(2) - 2.5 \\ &= 2 - 2.5 = -0.5 \end{aligned}$$

- (c) (3 pts) $k(1.5)$

$$\boxed{1.5}$$

$$\begin{aligned} k(1.5) &= m(1.5 + 1.5) \\ &= m(3) = 1.5 \end{aligned}$$

- (d) (3 pts) $n(-2.3)$

$$\boxed{0}$$

$$n(-2.3) = m(-2.3 + 2) = m(-0.3) = 0$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $-3, 7$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $-2, 10$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $0, 3$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$ 6

(b) (4 pts) $(f - g)(2)$

-2

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

0

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Harrington, Michael

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

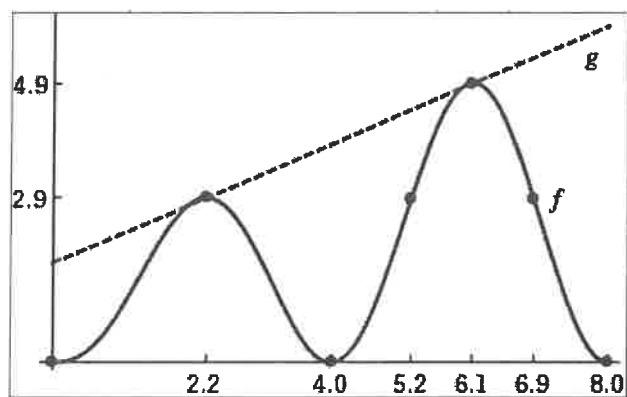
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = \boxed{2.9}$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers. *There is more than 1 value*

$x = f(x) = 0$ $[0, 4.0, 8.0]$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = f(x) = g(x)$ $[2.2, 2.9] [6.1, 4.9]$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

AV $= \frac{f(6.9) - f(4.0)}{6.9 - 4.0}$

$= \frac{2.9 - 0}{2.9} = 1$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$\begin{array}{l} 17-2x \\ 5-2x \\ x \end{array} \quad \left(\begin{array}{l} 17x-2x \\ 5x-2x \\ x \end{array} \right) \quad [P=f(t) \ 17+5]$$

$$(V) = 85$$

$$[V(x) = x(17-2x)(5-2x)]$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$(17, 5)$$

$$(a, b)$$

$$5-2x > 0 = x < \frac{5}{2}$$

$$x > 0$$

$$\left[0, \frac{5}{2} \right]$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

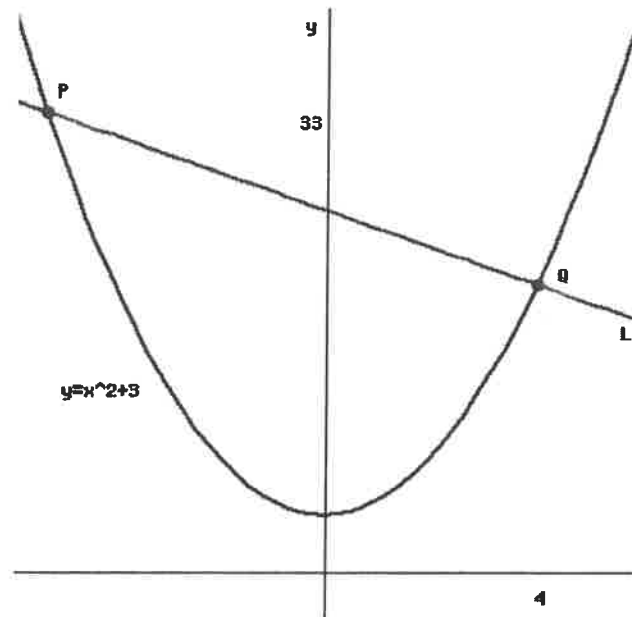
$$f(1) = 36 - (1)^2 = 36 - 1 = 36 \times 1 = 35$$

$$f(5) = 36 - (5)^2 = 36 - 25 = 11$$

$$\frac{f(5) - f(1)}{5 - 1}$$

$$\frac{11 - 35}{4} = \frac{-24}{4} = [-6]$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-5, 33)$$

$$Q = (4, 22)$$

$$y = 4^2 + 3 = y = 16 + 3 = y$$

$$y = 33^2 + 3 = 1089 + 3 = 1092$$

$$y = q(x) = ax^2 + bx + c$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$P + Q = L$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$x = \frac{-5}{2(1)} = -\frac{5}{2}$$

$$y = q(x) = ax^2 + bx + c$$

$$f(x) = x^2 + 5x - 14$$

$$f\left(-\frac{5}{2}\right) = \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14$$

$$\frac{25}{4} - \frac{25}{2} - 14 = \frac{25 - 50 - 56}{4} = -\frac{81}{4}$$

$$f(x) = x^2 + 5x - 14$$

$$x = \frac{-b}{2a}$$

$$\left[-2.5, -20.25\right] \left[-\frac{5}{2}, -\frac{81}{4}\right]$$

- (b) (2 pts) Determine the y -intercept of the parabola. $x=0$

$$y = (0, -14)$$

$$f(0) = 0^2 + 5(0) - 14 = -14$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

[There are none]

$$y = 3(x - 5)^2 - 9$$

$$y = 3(5 - 5)^2 - 9$$

$$3 \cdot 0^2 - 9$$

$$-9$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$(f \circ g)(3) = [9] \quad \frac{f(x) = 2(3) - 7}{3 + 5} = \frac{-1}{8}$$

(b) (3 pts) $(g \circ f)(3)$

$$(g \circ f)(3) = [4] \quad \frac{3 + 5}{2(3) - 7} \quad \frac{2(3) - 7}{2(3) - 7}$$

(c) (3 pts) $(f \circ f)(3)$

$$(f \circ f)(3) = [-4]$$

(d) (3 pts) $(g \circ g)(3)$

$$(g \circ g)(3) = [13]$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = [1]$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2)) = [2]$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(g(x)) = 6$$

$$g(x) = 1 \Rightarrow x = 4$$

$$g(x) = 3 \Rightarrow x = 0$$

$$[x = 0, 4]$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

$$C = g(x) = 30x + 597$$

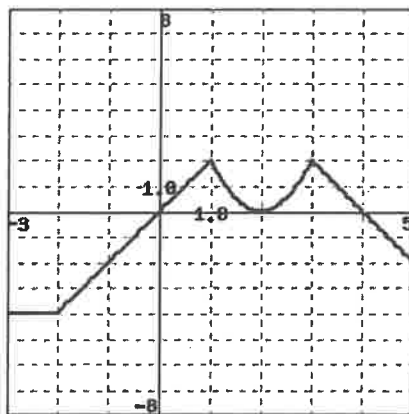
$$C = 30x + 597$$

$$C - 597 = 30x$$

$$\boxed{g^{-1}(C) = \frac{C - 597}{30}}$$

$$-1.9, 1$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1) = m(1) = -1$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5 = m(2) - 2.5 = -2.5$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) = m(3) = -2$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = -4 + 2 = -2$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

$$\underline{f(7) - f(-5)}$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) = (-3)^2 - 5 = [4]$$

$$(f \cdot g)(3)$$

$$\begin{aligned} f(x) &= x^2 - 5 \\ g(x) &= 4x + 1 \end{aligned}$$

(b) (4 pts) $(f - g)(2)$

$$(f - g)(2) = f(2) - g(2) = -1 - 4 = [-5]$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\left(\frac{f}{g}\right)(0) = \frac{f(0)}{g(0)} = \frac{-5}{1} = [-5]$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Marry Rose, WA3183813

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
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Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

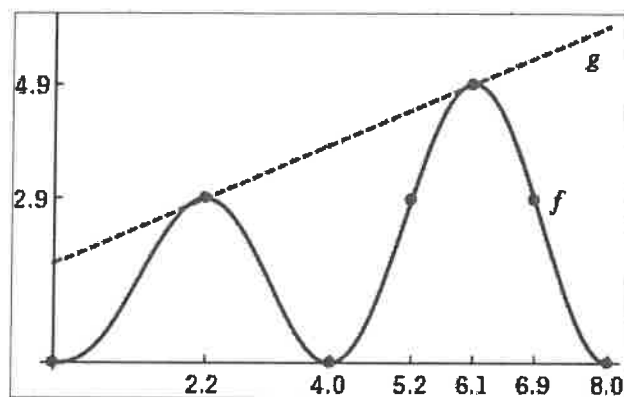
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 2.2, 5.2, 6.9$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 2.2, 6.1$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$f(4) = 0$$

$$f(6.9) = 2.9$$

$$AV_{[4, 6.9]} = \frac{f(6.9) - f(4)}{6.9 - 4}$$

$$= \frac{2.9 - 0}{2.9}$$

$$= -\frac{1.1}{2.9}$$

$$\approx 0.379$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

material 17 inches long 5 inches wide

$$\text{height} = x$$

$$\text{length} = 17 - 2x$$

$$\text{width} = 5 - 2x$$

$$V_{\text{(volume of } x)} = x(17 - 2x)(5 - 2x)$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$\text{For } x > 0$$

$$5 - 2x > 0$$

$$x < \frac{5}{2}$$

$$\text{Therefore } (0, \frac{5}{2})$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

$$f(1) = 36 - (1)^2 = 35$$

$$f(5) = 36 - (5)^2 = 36 - 25 = 11$$

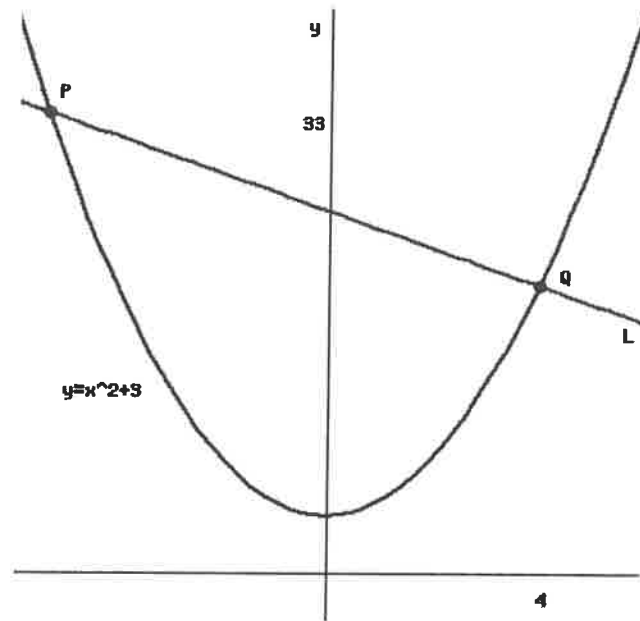
$$AV_{(1,5)} = \frac{f(5) - f(1)}{5 - 1}$$

$$= \frac{11 - 35}{4}$$

$$= -\frac{24}{4}$$

$$= -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$P = (-\sqrt{30}, 33)$ Since P is on the left side of the graph

$Q = (4, 19)$

For Q , $y = x^2 + 3$

$= (4)^2 + 3 = 16 + 3 = 19$

For P , $y = x^2 + 3$

$33 = x^2 + 3$

$x^2 = 30$

$x = \pm \sqrt{30}$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$(-\sqrt{30}, 33)$ $(4, 19)$
 $x_1 \quad y_1 \quad x_2 \quad y_2$

The slope of line (L) $= \frac{19 - 33}{4 - (-\sqrt{30})}$

$= \frac{-14}{4 + \sqrt{30}}$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$y - 19 = -\frac{14}{4 + \sqrt{30}}(x - 4)$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$f(x) = x^2 + 5x - 14$$

$$\text{Let } a = 1, b = 5$$

$$\text{Vertex formula, } -\frac{b}{2a}$$

$$= -\frac{5}{2(1)}$$

$$= -\frac{5}{2}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$\text{For } y\text{-intercept, } f(x) = 0$$

$$f(0) = (0)^2 + 5(0) - 14$$

$$f(0) = -14$$

$$\text{So: } (0, -14)$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 3(x - 5)^2 - 9$$

To find the zeros, set $y = 0$

$$3(x - 5)^2 - 9 = 0$$

$$3(x - 5)^2 = 9$$

$$(x - 5)^2 = \frac{9}{3}$$

$$(x - 5)^2 = 3$$

$$x - 5 = \pm\sqrt{3}$$

$$x = +5 \pm \sqrt{3}$$

$\therefore$ The zeros are $(+5 + \sqrt{3}), (+5, -\sqrt{3})$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$\begin{aligned}(f \circ g)(3) &= f(g(3)) \\ &= f(3+5) \\ &= f(8)\end{aligned}$$

$$\begin{aligned}(f \circ g)(3) &= 2(8) - 7 \\ &= 16 - 7 \\ &= 9 \quad \# \text{ Answer}\end{aligned}$$

(b) (3 pts) $(g \circ f)(3)$

$$\begin{aligned}(g \circ f)(3) &= g(f(3)) \\ &= g(2(3) - 7) \\ &= g(-1)\end{aligned}$$

$$\begin{aligned}(g \circ f)(3) &= -1 + 5 \\ &= 4 \quad \# \end{aligned}$$

(c) (3 pts) $(f \circ f)(3)$

$$\begin{aligned}(f \circ f)(3) &= f(f(3)) \\ &= f(2(3) - 7) \\ &= f(-1)\end{aligned}$$

$$\begin{aligned}(f \circ f)(3) &= 2(-1) - 7 \\ &= -2 - 7 \\ &= -9 \quad \# \end{aligned}$$

(d) (3 pts) $(g \circ g)(3)$

$$\begin{aligned}(g \circ g)(3) &= g(g(3)) \\ &= g(3+5) \\ &= g(8)\end{aligned}$$

$$\begin{aligned}(g \circ g)(3) &= 8 + 5 \\ &= 13 \quad \# \end{aligned}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$\begin{aligned}g^{-1}(0) &= g(1) \\ g(1) &= 0\end{aligned}$$

$$\text{So, } g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$\begin{aligned}g^{-1}(2) &= 2 \\ g(g^{-1}(2)) &= g(2)\end{aligned}$$

$$g(2) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(0) = 4, \quad f(4) = 1$$

$$g(2) = 0, \quad g(x) = 4$$

From the table,

$$g(1) = 0, \quad g(3) = 4$$

The value of x is $(1, 3)$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{C - 597}{30}$$

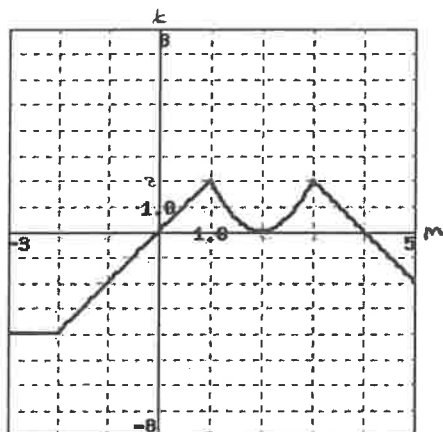
$$C = g(x) = 30x + 597$$

$$C = 30x + 597$$

$$C - 597 = 30x$$

$$x = \frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2) = m(2 - 1)$
 $= m(1), m(1) = 0$
 $p(2) = 0$
- (b) (3 pts) $w(2.5) = m(2.5 - 0.5)$
 $= m(2), m(2) = 0$
 $w(2.5) = 0(2) = 0$
- (c) (3 pts) $k(1.5) = m(1.5 + 1.5)$
 $= m(3), m(3) = 0$
 $k(1.5) = 0(3) = 0$
- (d) (3 pts) $n(-2.3) = m(-2.3) + 2$
 $m(-2.3) = -4$
 $n(-2.3) = -4 + 2$
 $n(-2.3) = -2$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-7, 7)$

$$-5 - 2 = -7$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

$$3 + 7 = 10$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(4, -8)$

$$-5 + 5 = 0$$

$$-4 - 4 = -8$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) = (-3)^2 - 5 = 9 - 5 = 4$$

$$g(-3) = 4(-3) + 1 = -12 + 1 = -11$$

$$\begin{aligned} (f \cdot g)(-3) &= f(-3) \cdot g(-3) \\ &= 4 \times (-11) \\ &= -44 \end{aligned}$$

(b) (4 pts) $(f - g)(2)$

$$f(2) = (2)^2 - 5 = 4 - 5 = -1$$

$$g(2) = 4(2) + 1 = 8 + 1 = 9$$

$$\begin{aligned} (f - g)(2) &= f(2) - g(2) \\ &= -1 - 9 \\ &= -10 \end{aligned}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\begin{aligned} f(0) &= (0)^2 - 5 \\ &= -5 \end{aligned}$$

$$\begin{aligned} g(0) &= 4(0) + 1 \\ &= 1 \end{aligned}$$

$$\left(\frac{f}{g}\right)(0) = \frac{-5}{1} = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Penelope Luna WA3000603

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

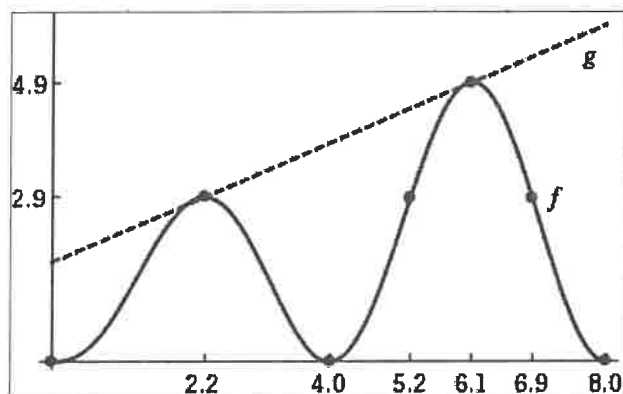
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 4.0, 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$f(6.9) - f(4.0)$

$f(4.0) = 0$ $f(6.9) =$
 $(4.0, 0)$ $(6.9, 4.9)$

$\frac{4.9 - 0}{6.9 - 4.0} = \frac{4.9}{2.9} = 1.689$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$17 - x - x = 17 - 2x$$

$$5 - x - x = 5 - 2x$$

$$V(x) = x(17 - 2x)(5 - 2x)$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$5 - 2x > 0$$

$$\frac{-2x}{-2} > \frac{-5}{-2} = x < 2.5$$

$$0 < x < 2.5$$

$$0, 2.5$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(1) = 36 - 1^2$$

$$f(1) = 35$$

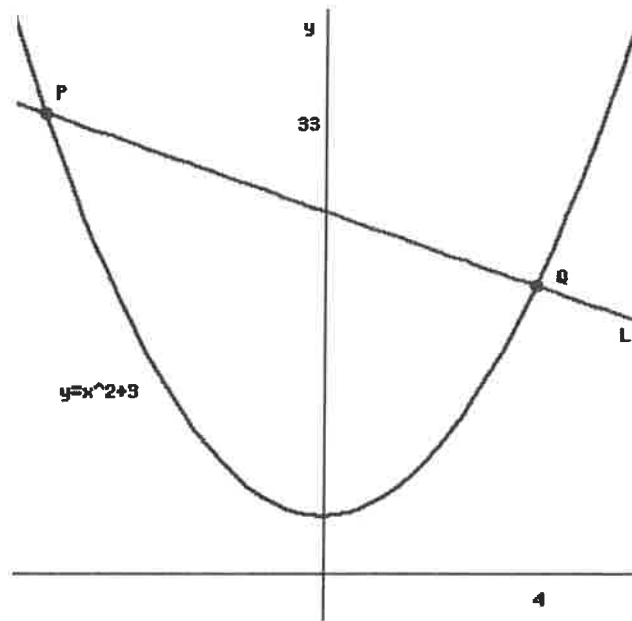
$$f(5) = 36 - 5^2$$

$$f(5) = 11$$

$$\frac{f(5) - f(1)}{5 - 1}$$

$$\frac{11 - 35}{4} = -\frac{24}{4} = \boxed{-6}$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-30, 33)$$

$$Q = (4, 19)$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = 4 - \sqrt{30}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y - 33 = (4 - \sqrt{30})(x + \sqrt{30})$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$\left(-\frac{5}{2}, -\frac{81}{4}\right)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$(0, -14)$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$0 = 3(x - 5)^2 - 9$$

$$9 = 3(x - 5)^2$$

$$3 = (x - 5)^2$$

$$x - 5 = \pm \sqrt{3}$$

$$x - 5 = \sqrt{3}$$

$$x - 5 = -\sqrt{3}$$

$$x = 5 \pm \sqrt{3}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3) = 9$

$$g(3) = 3 + 5 = 8$$

$$f(8) = 2(8) - 7 = 16 - 7 = 9$$

(b) (3 pts) $(g \circ f)(3) = 4$

$$f(3) = 2(3) - 7 = 6 - 7 = -1$$

$$g(-1) = -1 + 5 = 4$$

(c) (3 pts) $(f \circ f)(3) = -9$

$$f(3) = 2(3) - 7 = -1$$

$$f(-1) = 2(-1) - 7 = -9$$

(d) (3 pts) $(g \circ g)(3) = 13$

$$g(3) = 3 + 5 = 8$$

$$g(8) = 8 + 5 = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g^{-1}(2) = 2$$

$$g(2) = 1$$

$$g(g^{-1}(2)) = 1$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(1) = 6 \quad f(3) = 6$$

$$g(x) = 1$$

$$x = 4$$

$$g(x) = 3$$

$$x = 0$$

$$x = 4, 0$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

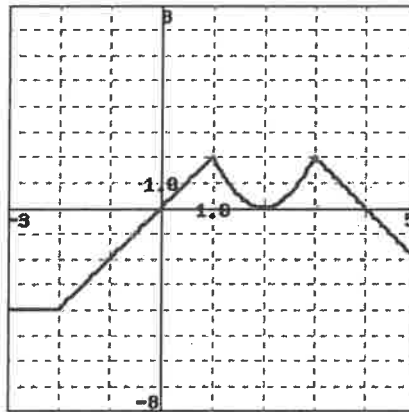
$$g^{-1}(C) = \frac{C - 597}{30}$$

$$C = 597 + 30x$$

$$\frac{C - 597}{30} = \frac{30x}{30}$$

$$\frac{C - 597}{30} = x$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2) = 2$

$$p(2) = m(2 - 1)$$

$$p = m(1)$$

(b) (3 pts) $w(2.5) = -0.5$

$$w(2.5) = m(2.5 - 0.5) - 2.5 = m(2) - 2.5 = -1 + -0.5 = -1.5$$

(c) (3 pts) $k(1.5) = 2$

$$k(1.5) = m(1.5 + 1.5)$$

$$k(1.5) = m(3)$$

(d) (3 pts) $n(-2.3) = -0.8$

$$n(-2.3) = m(-2.3) + 2$$

$$n = (-0.3)$$

$$-0.5 + -0.3 = -0.8$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $\boxed{(-3, 7)}$

moved 2 units to the right

$$-5 + 2 = -3$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $\boxed{(-5, 10)}$

moves up 3 units

$$7 + 3 = 10$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $\boxed{(-7, 11)}$

$f(x + 5) \rightarrow$ left 5 units

$-4 \rightarrow$ move down

$$-5 - 2 = -7$$

$$7 - 4 = 3$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $\boxed{(f \cdot g)(-3) = 154}$

$$f(-3) = (-3)^2 - 5$$

$$f(-3) = 9 - 5 = 4$$

$$g(-3) = 4(-3) + 1$$

$$= -12 + 1 = -11$$

$$-14(-11) = 154$$

(b) (4 pts) $\boxed{(f - g)(2) = -10}$

$$f(2) = 2^2 - 5$$

$$= 4 - 5 = -1$$

$$g(2) = 4(2) + 1$$

$$= 8 + 1 = 9$$

$$-1 - 9 = -10$$

(c) (4 pts) $\boxed{\left(\frac{f}{g}\right)(0) = -5}$

$$\frac{f}{g}(0) = \frac{0^2 - 5}{1}$$

$$= -5$$

$$g(0) = 4(0) + 1 = 1$$

$$-\frac{5}{1} = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Garcia, Jocelyn. WA3267035

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$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

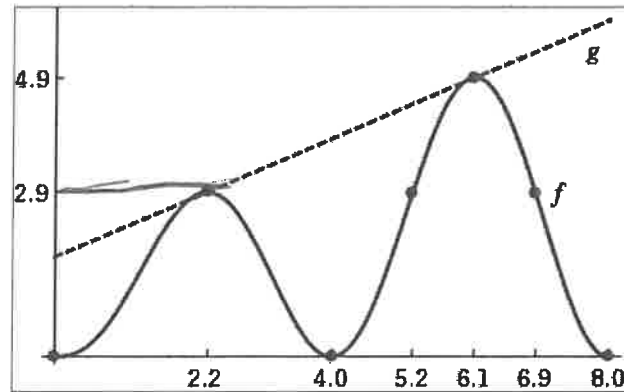
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) =$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 4.0, 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = 17 \cdot 5 \cdot x$$

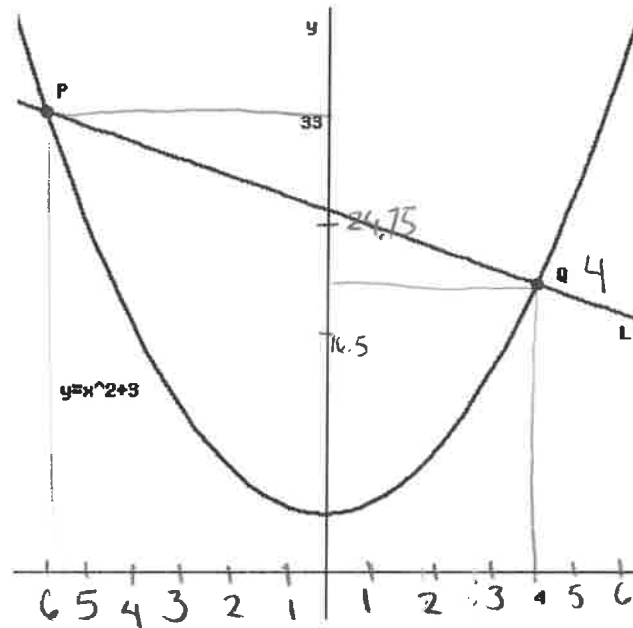
$$V = L \cdot W \cdot h$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[1,5]} = \frac{f(5) - f(1)}{5 - 1} = \frac{4}{4} = 1$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (6, 33)$$

$$Q = (4, 16.5)$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

- (b) (2 pts) Determine the y -intercept of the parabola.

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(3) + g(3) =$$

(b) (3 pts) $(g \circ f)(3)$

(c) (3 pts) $(f \circ f)(3)$

(d) (3 pts) $(g \circ g)(3)$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$4$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

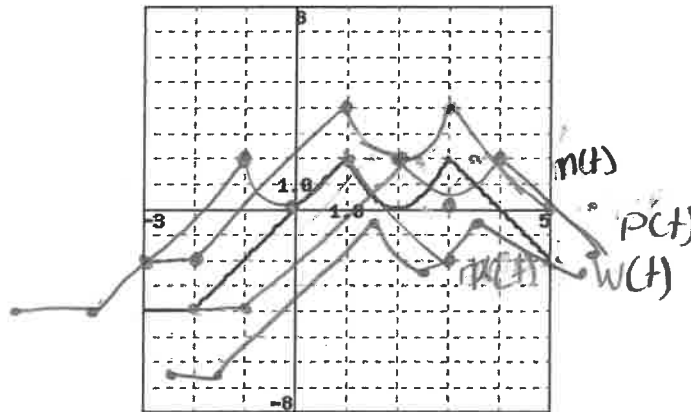
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$$x = 0$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = -30x - 597$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2)$

(b) (3 pts) $w(2.5)$

(c) (3 pts) $k(1.5)$

-0.5

(d) (3 pts) $n(-2.3)$

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12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$
 $f(-3) \cdot g(-3)$

(b) (4 pts) $(f - g)(2)$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Huang, David WA3278621

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$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

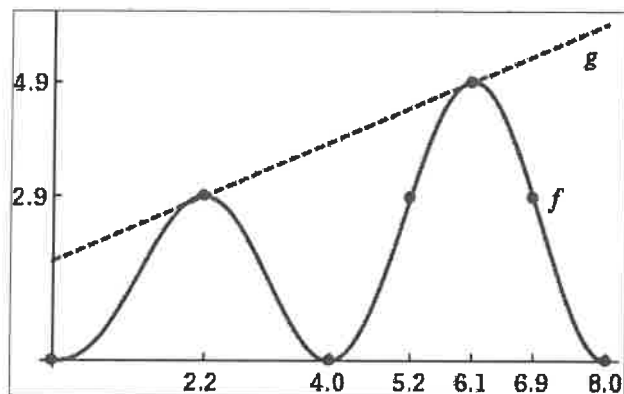
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

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- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 4.0, 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

AV[4.0, 6.9]

$$\frac{f(b) - f(a)}{b - a} = \frac{f(6.9) - f(4.0)}{6.9 - 4.0} = \frac{2.9 - 0}{2.9} = \frac{2.9}{2.9} = 1$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$\begin{aligned}\text{Length: } 17 - x - x &= 17 - 2x \\ \text{Width: } 5 - x - x &= 5 - 2x \\ V(x) &= x(17 - 2x)(5 - 2x)\end{aligned}$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

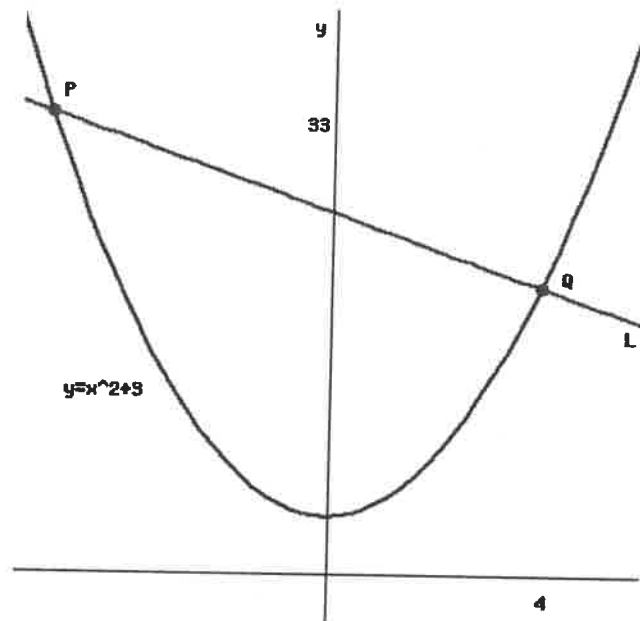
$$\begin{aligned}\text{Length: } 17 - 2x &> 0 \\ 17 &> 2x \\ 8.5 &> x \text{ or } x < 8.5\end{aligned}$$

$$\begin{aligned}\text{Width: } 5 - 2x &> 0 \\ 5 &> 2x \\ 2.5 &> x \text{ or } x < 2.5\end{aligned}$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$\begin{aligned}f(x) &= 36 - x^2 \\ [1, 5] \\ f(1) &= 36 - (1)^2 = 36 - 1 = 35 \\ f(5) &= 36 - (5)^2 = 36 - 25 = 11 \\ \text{Average rate of change} &= \frac{f(5) - f(1)}{5 - 1} = \frac{11 - 35}{4} = \frac{-24}{4} = -6\end{aligned}$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = 33 \quad y = 33; 3 = x^2 + 3 \rightarrow x^2 = 30 \quad x = \pm\sqrt{30} = (-\sqrt{30}, 33)$$

$$Q = 4$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$y = 4^2 + 3 = 19$$

$$Q = (4, 19)$$

$$(19, 33) / (4 - (-\sqrt{30}))$$

$$= 14 / (4 + \sqrt{30}) = 4 - \sqrt{30}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = (4 - \sqrt{30})x + (3 + 4\sqrt{30})$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$f(x) = x^2 + 5x - 14 \quad \text{vertex: } x = -b/(2a) \\ = (-5)/2 = -2.5 \quad y = f(-2.5) = 7.75$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$\begin{array}{c} -14 \\ (-7, 2) \end{array} \quad (x-7)(x+2) = 0 \rightarrow x = 7, -2 \quad x\text{-int: } 7, -2 \quad y\text{-int: } (0, -14)$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 3(x - 5)^2 - 9 \\ 3(x - 5)^2 - 9 = 0 \\ 3(x - 5)^2 - 9 \rightarrow (x - 5)^2 = 3 \\ x - 5 = \pm\sqrt{3} \\ x = -5 + \sqrt{2} \text{ or } x = -5 - \sqrt{2}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$g(3) = 3 + 5 = 8$$

$$f(8) = 2(8) - 7 = 9$$

(b) (3 pts) $(g \circ f)(3)$

$$f(3) = 2(3) - 7 = -1$$

$$g(-1) = -1 + 5 = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f(3) = 2(3) - 7 = -1$$

$$f(-1) = 2(-1) - 7 = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$g(3) = 3 + 5 = 8$$

$$g(8) = 8 + 5 = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g(x) = 0 \text{ when } x = 1 = g^{-1}(0) = 2$$

(b) (2 pts) Find $g(g^{-1}(2))$

An inverse undoes g so, $g(g^{-1}(2)) = 2$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(g(1)) = 6$$

$$f(g(2)) = 6$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

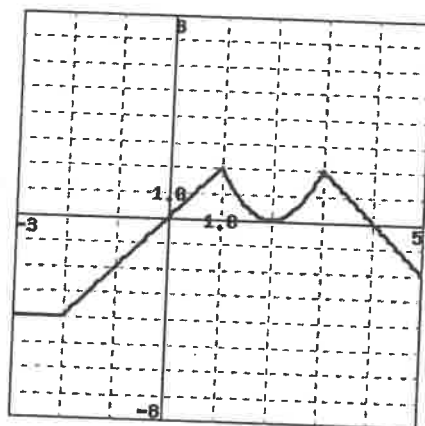
$$C = 30x + 597$$

$$C - 597 = 30x$$

$$g^{-1}(C) = (C - 597) / 30$$

$$= -19.9$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$
- (b) (3 pts) $w(2.5)$
- (c) (3 pts) $k(1.5)$
- (d) (3 pts) $n(-2.3)$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $-3, 7$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $-5, 10$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $-10, 3$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$ $(f \cdot g)(-3) = f(-3) \cdot g(-3) = -3 \cdot -12 = 36$

(b) (4 pts) $(f - g)(2)$ $f(2) - g(2) = 2 - 8 = -6$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\frac{f(0)}{g(0)} = 0$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Lo, Cayla

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and **box all final answers**.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

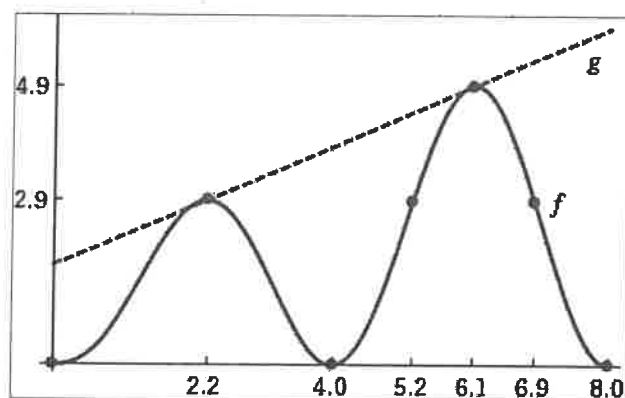
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = \boxed{2.9}$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = \boxed{0, 4.0, 8.0}$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = \boxed{2.2, 6.1}$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$AV_{[4.0, 6.9]} = \frac{f(6.9) - f(4.0)}{6.9 - 4.0}$$

$$= \frac{2.9 - 0}{6.9 - 4.0}$$

$$= \boxed{1}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V(x) = (17 - 2x)(5 - 2x) \cdot x$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$0 = 17 - 2x$$

$$17 = 2x$$

$$\frac{17}{2} = x$$

$$x = 8.5$$

$$0 = 5 - 2x$$

$$5 = 2x$$

$$\frac{5}{2} = x$$

$$x = 2.5$$

$$\text{domain: } (0, 2.5)$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[1,5]} = \frac{f(5) - f(1)}{5 - 1}$$

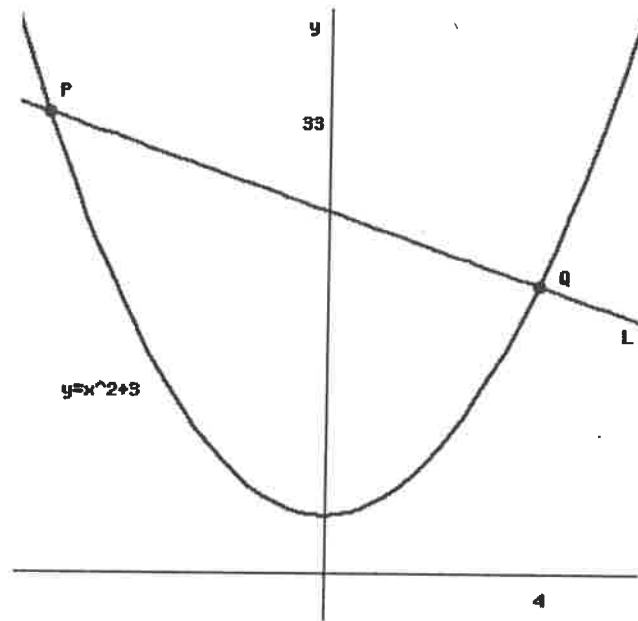
$$= \frac{11 - 35}{4}$$

$$= -6$$

$$f(5) = 36 - 5^2$$
$$= 11$$

$$f(1) = 36 - 1^2$$
$$= 35$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\text{slope} = \frac{19 - 33}{4 - (-\sqrt{30})}$$

$$= \boxed{4 - \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$L = 19 + (4 - \sqrt{30})(x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)$$

$$\begin{aligned} &-\frac{5}{2(1)} \\ &= -\frac{5}{2} \\ &= -2.5 \end{aligned}$$

$$\begin{aligned} f\left(-\frac{5}{2}\right) &= \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14 \\ &= -\frac{81}{4} \\ &= -20.25 \end{aligned}$$

$$\boxed{\text{Vertex} = (-2.5, -20.25)}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$\boxed{y\text{-intercept} = -14}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\begin{aligned} &3(x-5)^2 - 9 \\ &= 3(x-5)(x-5) - 9 \\ &= 3(x^2 - 5x - 5x + 25) - 9 \\ &= 3(x^2 - 10x + 25) - 9 \\ &= 3x^2 - 30x + 75 - 9 \\ &= 3x^2 - 30x + 66 \end{aligned}$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{-30 \pm \sqrt{30^2 - 4(3)(66)}}{2(3)}$$

$$= -5 \pm \sqrt{3}$$

$$\frac{-30 - \sqrt{30^2 - 4(3)(66)}}{2(3)}$$

$$= -5 - \sqrt{3}$$

$$\boxed{x = -5 + \sqrt{3}, -5 - \sqrt{3}}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$(f \circ g)(3) = f(8) \\ = \boxed{9}$$

$$g(3) = 3 + 5 \\ = 8$$

$$f(8) = 2(8) - 7$$

$$= 16 - 7 \\ = 9$$

$$g(-1) = -1 \\ = 4$$

(b) (3 pts) $(g \circ f)(3)$

$$(g \circ f)(3) = g(1) \\ = \boxed{4}$$

$$f(3) = 2(3) - 7 \\ = 6 - 7 \\ = -1$$

(c) (3 pts) $(f \circ f)(3)$

$$(f \circ f)(3) = f(-1) \\ = \boxed{-9}$$

$$f(-1) = 2(-1) - 7 \\ = -2 - 7 \\ = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$(g \circ g)(3) = g(8) \\ = \boxed{13}$$

$$g(8) = 8 + 5 \\ = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$\boxed{3}$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$\boxed{2}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(1) = 6$$

$$f(3) = 6$$

$$g(4) = 1$$

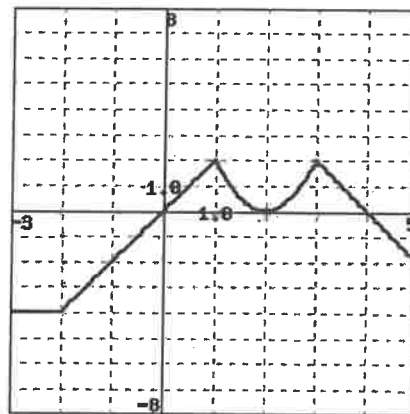
$$g(0) = 3$$

$$\boxed{x = 4, 0}$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2)$ $p(2) = m(2 - 1) = m(1) = \boxed{2}$

(b) (3 pts) $w(2.5)$ $w(2.5) = m(2.5 - 0.5) - 2.5 = m(2) - 2.5 = \boxed{0}$

(c) (3 pts) $k(1.5)$ $k(1.5) = m(1.5 + 1.5) = m(3) = \boxed{2}$

(d) (3 pts) $n(-2.3)$ $n(-2.3) = m(-2.3) + 2 = (-4) + 2 = \boxed{-2}$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$\begin{aligned}(f \cdot g)(-3) &= ((-3)^2 - 5)(4(-3) + 1) = \boxed{-44} \\ &= (9 - 5)(-12 + 1) \\ &= (4)(-11)\end{aligned}$$

(b) (4 pts) $(f - g)(2)$

$$\begin{aligned}(f - g)(2) &= ((2)^2 - 5) - (4(2) + 1) = \boxed{-10} \\ &= (4 - 5) - (8 + 1)\end{aligned}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\left(\frac{f}{g}\right)(0) = \frac{(0)^2 - 5}{4(0) + 1}$$

$$= \frac{-5}{1}$$

$$= \boxed{-5}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: ENGALAN, STEPHANEE, S00017399

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Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

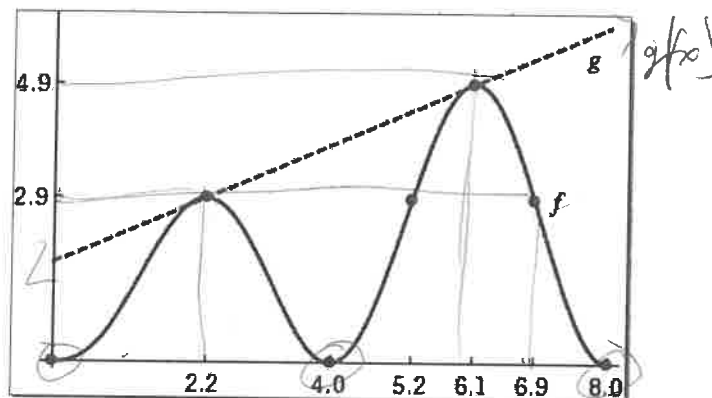
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = (0, 4.0, 8.0)$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = (2.2, 6.1)$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a} \quad (4.0, 0) \quad (6.9, 2.9)$$

$$AV_{[4.0, 6.9]} = \frac{2.9 - 0}{6.9 - 4.0}$$

$$= \frac{2.9}{2.9} \text{ or simply } 1$$



2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = L \cdot W \cdot H$$

$$L = 17 - x - x$$

$$L = 17 - 2x$$

$$W = 5 - x - x$$

$$W = 5 - 2x$$

$$V = L \cdot W \cdot H$$

$$V = (17 - 2x)(5 - 2x)(x)$$

$$V = x(17 - 2x)(5 - 2x)$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$L = 17 - 2x > 0 \rightarrow x < 8.5$$

$$W = 5 - 2x > 0 \rightarrow x < 2.5$$

$$H = x > 0$$

To get the domain

$$5 - 2(2.5) = 0$$

Therefore the domain is

$$(0, 2.5)$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

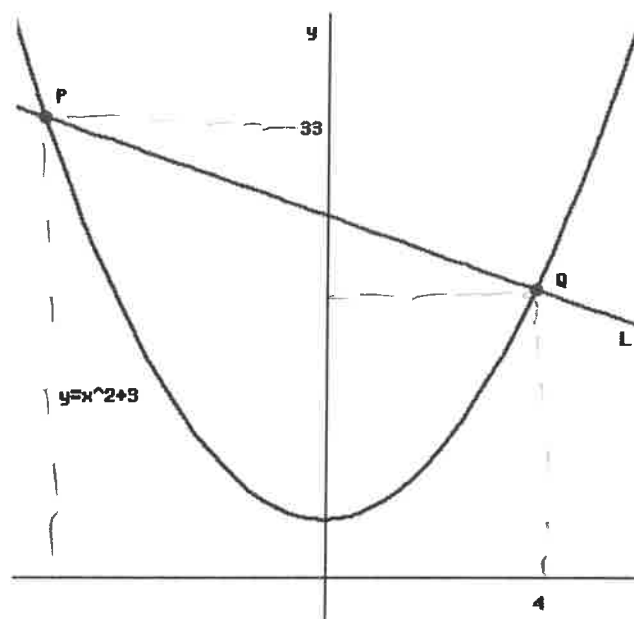
$$AV_{[1,5]} = \frac{(36 - (5)^2) - (36 - (1)^2)}{5 - 1}$$

$$= \frac{11 - 35}{4}$$

$$= \frac{-24}{4}$$

$$= -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

isolate x

$$P = (x, 33)$$

$$y = x^2 + 3$$

$$33 = x^2 + 3$$

$$-3$$

$$\sqrt{30} = \sqrt{30}$$

$$x = \pm\sqrt{30}$$

Because P is in the left side of x axis then the x is negative therefore $x = -\sqrt{30}$

$$Q = (4, y)$$

$$y = x^2 + 3$$

$$y = (4)^2 + 3$$

$$y = 19$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

$$m = \frac{19 - 33}{4 - (-\sqrt{30})}$$

$$m = \frac{-14}{4 + \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

Point slope form

$$y = y_0 + m(x - x_0)$$

point slope form of $Q (4, 19)$

$$L = 19 + \frac{-14}{4 + \sqrt{30}}(x - 4)$$

Vertex form is $= a(x-h)^2 + k$ | Vertex formula for x-coordinate $\rightarrow x = \frac{-b}{2a}$
 Vertex $= V(h, k)$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

standard form $\leftarrow f(x) = x^2 + 5x - 14$
 $\begin{matrix} a & b & c \end{matrix}$

- (a) (6 pts) Determine the vertex of the parabola.

Find the x in vertex

$$x = \frac{-b}{2a}$$

$$x = \frac{-5}{2(1)}$$

$$x = \frac{-5}{2}$$

$$x = -2.5$$

Find the y (use the quad funct.)

$$f(x) = x^2 + 5x - 14$$

$$f(-2.5) = (-2.5)^2 + 5(-2.5) - 14$$

$$= 6.25 + 12.5 - 14$$

$$f(-2.5) = 4.75$$

Therefore the vertex point of the parabola is

$$(-2.5, 4.75)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

sub $x=0$ in the function

$$f(x) = x^2 + 5x - 14$$

$$f(0) = (0)^2 + 5(0) - 14$$

$$= -14 \rightarrow y\text{-intercept}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x-5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

Assume the function is $= 0$

$$0 = 3(x-5)^2 - 9$$

Isolate the squared form

$$\begin{matrix} 0 & = & 3(x-5)^2 & - & 9 \\ +9 & & +9 & & \end{matrix}$$

$$\frac{9}{3} = \frac{3(x-5)^2}{3}$$

$$\frac{9}{3} = (x-5)^2 \text{ or } 3 = (x-5)^2$$

Now isolate the x

$$\sqrt{3} = \sqrt{(x-5)^2}$$

$$\pm\sqrt{3} + 5 = x$$

$$x = \pm\sqrt{3} + 5$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3) = 2x - 7$
 $= [2(x+5) - 7]$
 $= [2(3+5) - 7]$
 $= [2(8) - 7]$
 $= 16 - 7$
 $= 9$

(b) (3 pts) $(g \circ f)(3) = x + 5$
 $= [(2x - 7) + 5]$
 $= [(2(3) - 7) + 5]$
 $= [(6 - 7) + 5]$
 $= -1 + 5$
 $= 4$

(c) (3 pts) $(f \circ f)(3) = 2x - 7$
 $= [2(2x - 7) - 7]$
 $= [2(2(3) - 7) - 7]$
 $= [2(6 - 7) - 7]$
 $= [2(-1) - 7]$
 $= -2 - 7$
 $= -9$

(d) (3 pts) $(g \circ g)(3) = x + 5$
 $= [(x + 5) + 5]$
 $= [(3) + 5] + 5$
 $= 8 + 5$
 $= 13$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$
 $g^{-1}(0) = 1$

(b) (2 pts) Find $g(g^{-1}(2))$
 $g(g^{-1}(2)) = 2$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$f(g(x)) = 6$

$f(1)$ and $f(3) = 6$

What is the x of $g(x) = 1$ and 3

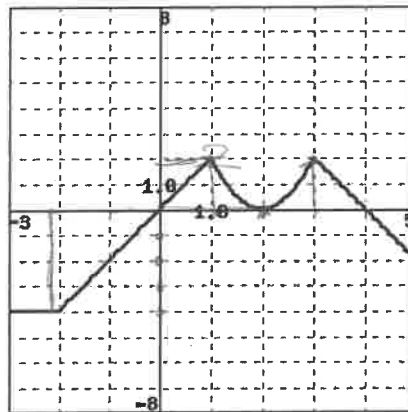
$x = 0$ and $x = 4$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

$$\begin{aligned} C &= 30x + 597 \\ -597 & \quad -597 \\ \hline C - 597 &= 30x \\ \frac{C - 597}{30} &= x \\ \boxed{y = \frac{C - 597}{30}} \end{aligned}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2) = m(t - 1) \rightarrow$ in graph the output of 1 is $\boxed{= 2}$
 $= m(2 - 1)$
 $= 1$
- (b) (3 pts) $w(2.5) = m(t - 0.5) - 2.5 \rightarrow = 0 - 2.5$
 $= m(2.5 - 0.5) - 2.5$
 $= (2) - 2.5$
 $\boxed{= -2.5}$
- (c) (3 pts) $k(1.5) = m(t + 1.5) \rightarrow \boxed{= 2}$ output of 3 is 2
 $= m(1.5 + 1.5)$
 $= 3$
- (d) (3 pts) $n(-2.3) = m(t) + 2 \rightarrow$ output of -2.3 is -4
 $= m(-2.3) + 2$
 $= -4 + 2$
 $\boxed{= -2}$

Vertical translation $\rightarrow g(x) = f(x) + a$
 Horizontal translation $\rightarrow h(x) = f(x - b)$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-5 + 2, 7) \rightarrow (-3, 7)$

$\hookrightarrow$ Horizontal translation
 meaning the graph move to the right by 2 units

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 7 + 3) \rightarrow (-5, 10)$

$\hookrightarrow$ vertical translation
 meaning the graph moves upward by 3 units

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-5 - 5, 7 - 4) \rightarrow (-10, 3)$

$\hookrightarrow$ Both translation
 meaning the graph move to the left by 5 units and move downward by 4 units

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) = (x^2 - 5) \cdot (4x + 1)$
 $= [(-3)^2 - 5] \cdot [4(-3) + 1]$
 $= (9 - 5) \cdot (-12 + 1)$
 $= 4 \cdot (-11)$
 $= -44$

(b) (4 pts) $(f - g)(2) = (x^2 - 5) - (4x + 1)$
 $= [(2)^2 - 5] - [4(2) + 1]$
 $= (4 - 5) - (8 + 1)$
 $= -1 - 9$
 $= -10$

(c) (4 pts) $\left(\frac{f}{g}\right)(0) = \frac{x^2 - 5}{4x + 1}$
 $= \frac{(0)^2 - 5}{4(0) + 1}$
 $= \frac{-5}{1}$
 $= -5$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: MYA T, JEFFEAY, WA3175177

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

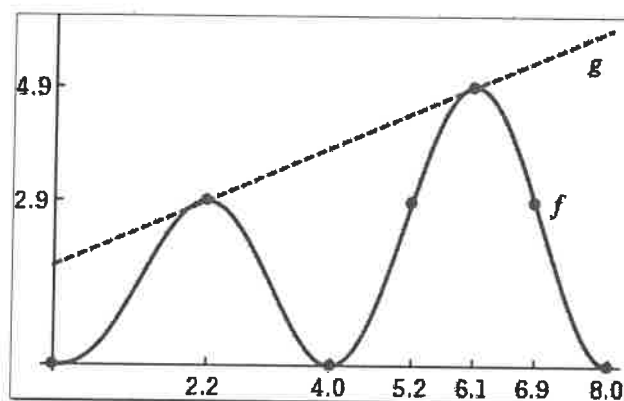
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 4.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 4.0$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 2.2, 6.1$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$(4.0, 0) \quad (6.9, 2.9)$$

$$f(4, 6.9) = \frac{2.9 - 0}{6.9 - 4.0} = \frac{2.9}{2.9} = 1$$

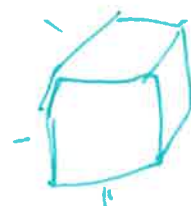
2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

Let the height be x
length = $x^2 - 17$
width = $x^2 - 5$

$$V = \text{length} \times \text{width} \times \text{height}$$

$$= (x^2 - 17)(x^2 - 5)x$$



- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$x - 5 > 0$$

$$x > 5$$

$$0 < x < 5$$

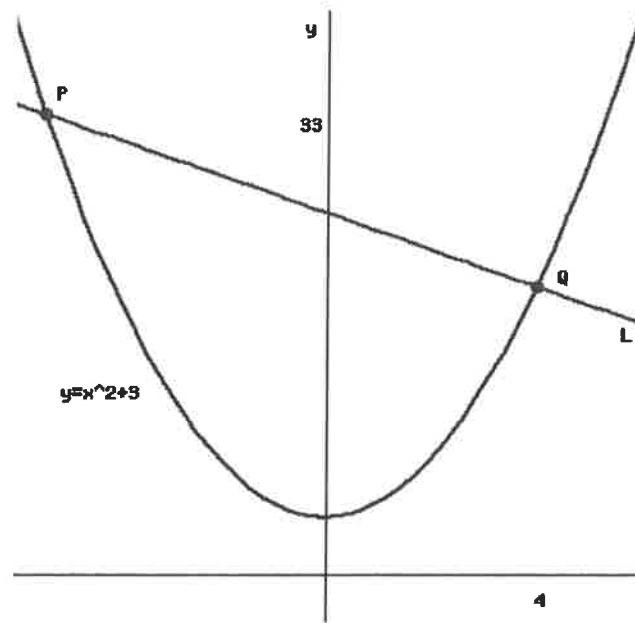
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(1) = 36 - 1^2 = 35$$

$$f(5) = 36 - 25 = 11$$

$$\frac{f(5) - f(1)}{5 - 1} = \frac{11 - 35}{4} = \frac{-24}{4} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$P = x^2 + 3, 33 = x^2 + 3, 30 = x^2, \pm\sqrt{30} = x, x = \sqrt{30}, x = -\sqrt{30}$
 on the left side
 so $x = -\sqrt{30}$
 $Q = 19$

$P(-\sqrt{30}, -\sqrt{30}), Q(4, 19)$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$Q(4, 19), P(-\sqrt{30}, -\sqrt{30})$
 $y = y_0 + m(x - x_0)$
 $19 = -\sqrt{30} + m(4 - \sqrt{30})$
 $19 + \sqrt{30} = m(4 - \sqrt{30})$
 $m = \frac{19 + \sqrt{30}}{4 - \sqrt{30}}$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

$\begin{matrix} a & b & c \end{matrix}$

- (a) (6 pts) Determine the vertex of the parabola.

$$\text{vertex} = \frac{-b}{2a} = \frac{-5}{2 \times 1} = -\frac{5}{2}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$y = (-7, 0) (2, 0)$$

$$0 = x^2 + 5x - 14$$

$$0 = (x+7)(x-2)$$

$$f(2) = 2^2 + 5(2) - 14 = 4 + 10 - 14 = 0$$

$$\begin{array}{r} x \quad + \quad 7 \\ \times \quad - \quad 2 \\ \hline \end{array}$$

$$x = -7, x = 2$$

$$f(-7) = 49 - 35 - 14 = 0$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x-5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\text{lets } y \text{ be } 0$$

$$0 = 3(x-5)^2 - 9$$

$$9 = 3(x-5)^2$$

$$3 = (x-5)^2$$

$$\pm\sqrt{3} = x-5$$

$$x = 5 \pm \sqrt{3}$$

$$\text{so } x = 5 + \sqrt{3} \text{ or } x = 5 - \sqrt{3}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$ $\approx f(g(3)) \approx f(8)$
 $\approx 16 - 7$
 ≈ 9

(b) (3 pts) $(g \circ f)(3)$ $\approx g(f(3))$
 $\approx g(-1)$
 $\approx -1 + 5$
 ≈ 4

(c) (3 pts) $(f \circ f)(3)$ $\approx f(f(3)) \approx f(-1)$
 $\approx 2(-1) - 7$
 $\approx -2 - 7 = -9$

(d) (3 pts) $(g \circ g)(3)$ $\approx g(g(3)) \approx g(8)$
 $\approx 8 + 5 = 13$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$ ≈ 1

(b) (2 pts) Find $g(g^{-1}(2))$ ≈ 2

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$x \approx 4$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

$$C = 30x + 597$$

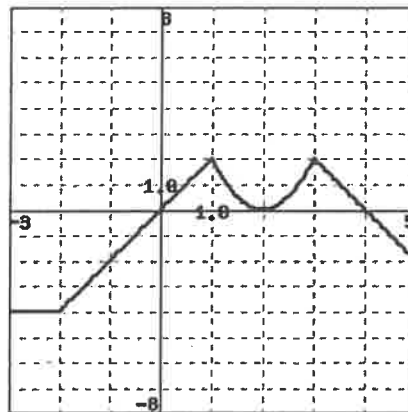
$$C - 597 = 30x$$

$$C - 597 = 30x$$

$$\frac{C - 597}{30} = x$$

$$g^{-1}(C) = \frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2) = m(2 - 1)$
 $= 2$

(b) (3 pts) $w(2.5) = m(2.5 - 0.5)$
 $= 0$

(c) (3 pts) $k(1.5) = m(1.5 + 1.5)$
 $=$

(d) (3 pts) $n(-2.3) = m(-2.3) + 2$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $-3, 7$

$$f(-5) = 7$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $-5, 10$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $1, 3$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) = f(-3) \cdot g(-3)$
 $= 4 \times (-11)$
 $= -44$

(b) (4 pts) $(f - g)(2) = f(2) - g(2)$
 $= 1 - (9)$
 $= -8$

(c) (4 pts) $\left(\frac{f}{g}\right)(0) = \frac{f(0)}{g(0)} = \frac{-5}{1} = -5$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Ryan Angel WA2995374

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Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

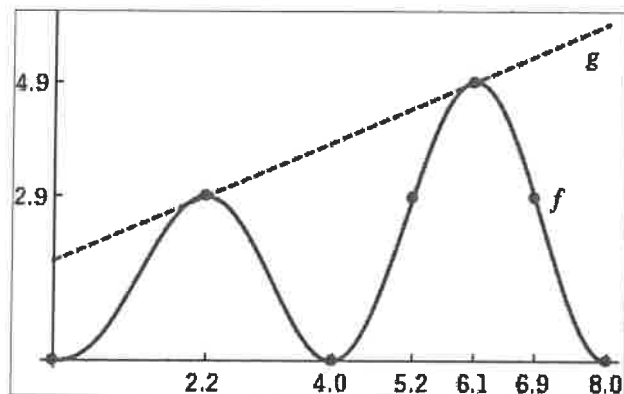
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 0, 4.0, 8.0$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 2.2, 6.1$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$? $\rightarrow (6.9, 2.9)$

$$\downarrow$$

$$(4.0, 0)$$

$$\frac{2.9 - 0}{6.9 - 4.0} = \frac{2.9}{2.9} = 1$$

the average rate of change of $f(x)$ on the interval

$$4.0 \leq x \leq 6.9 \text{ is } \frac{2.9}{2.9} \text{ which is } 1.$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = l \cdot w \cdot h$$

$$V = 17 - 5 - x$$

$$V = (x)(17 - 2x)(5 - 2x)$$

$$l = 17$$

$$17 - 2x$$

$$w = 5$$

$$5 - 2x$$

$$h = x$$

from cutting the

corners

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$17 - 2x > 0$$

$$5 - 2x > 0$$

$$x > 0$$

$$+2x \quad +2x$$

$$+2x \quad +2x$$

$$\frac{17}{2} > \frac{2x}{2}$$

$$\frac{5}{2} > \frac{2x}{2}$$

domain as an open

interval form: $(0, 2.5)$

$$8.5 > x$$

$$2.5 > x$$

← strictest restriction

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(1) = 36 - (1)^2$$

$$f(5) = 36 - (5)^2$$

$$\frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

$$f(1) = 36 - 1$$

$$f(5) = 36 - 25$$

$$f(1) = 35$$

$$f(5) = 11$$

$$(1, 35)$$

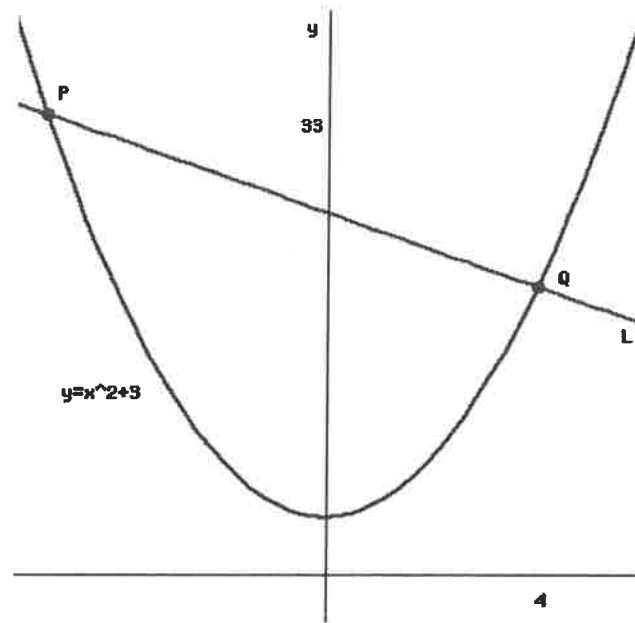
$$(5, 11)$$

the average rate of change

on the interval $[1, 5]$ is

-6.

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$\begin{array}{l}
 P = (-\sqrt{30}, 33) \\
 Q = (4, 19)
 \end{array}
 \quad
 \begin{array}{l}
 33 = x^2 + 3 \\
 -3 \quad -3 \\
 \sqrt{30} = \sqrt{x^2} \\
 x = \pm\sqrt{30}
 \end{array}
 \quad
 \begin{array}{l}
 y = (4)^2 + 3 \\
 y = 16 + 3 \\
 y = 19
 \end{array}$$

$\uparrow$ because the point is on the left, it can only be $-\sqrt{30}$ so $x = -\sqrt{30}$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\frac{19 - 33}{4 - (-\sqrt{30})} = \frac{-14}{4 + \sqrt{30}}$$

the slope of the line is $\frac{-14}{4 + \sqrt{30}}$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = 19 + \frac{-14}{4 + \sqrt{30}} (x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14. \quad a = 1 \quad b = 5 \quad c = -14$$

- (a) (6 pts) Determine the vertex of the parabola.

$$h = -\frac{b}{2a} \quad h = \frac{-5}{2(1)} \quad h = \frac{-5}{2} = \quad h = -2.5$$

$$k = f\left(-\frac{b}{2a}\right)$$

$$k = f(-2.5) = (-2.5)^2 + 5(-2.5) - 14 = f(-2.5) = 6.25 + -12.5 - 14$$

$$k = f(-2.5) = -20.25$$

$$\boxed{\text{Vertex} = (-2.5, -20.25)}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$\boxed{y \text{ int} = -14}$$

$$f(0) = (0)^2 + 5(0) - 14 = -14$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\begin{array}{ccc} 0 = 3(x-5)^2 - 9 & & \\ +9 & & +9 \end{array}$$

$$\boxed{x = \sqrt{3} + 5 \quad x = -\sqrt{3} + 5}$$

$$\begin{array}{ccc} 9 = 3(x-5)^2 & & \\ \hline 3 & & 3 \end{array}$$

$$\sqrt{3} = \sqrt{(x-5)^2}$$

$$\begin{array}{ccc} \pm \sqrt{3} = x - 5 & & \\ +5 & & +5 \end{array}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$g(3) = 3 + 5 \quad f(8) = 2(8) - 7 \quad \boxed{(f \circ g)(3) = 9}$$

$$g(3) = 8 \quad f(8) = 9$$

(b) (3 pts) $(g \circ f)(3)$

$$f(3) = 2(3) - 7 \quad g(-1) = (-1) + 5 \quad \boxed{(g \circ f)(3) = 4}$$

$$f(3) = -1 \quad g(-1) = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f(3) = -1 \quad \boxed{(f \circ f)(3) = -9}$$

$$f(-1) = 2(-1) - 7$$

(d) (3 pts) $(g \circ g)(3)$

$$g(3) = 8 \quad \boxed{(g \circ g)(3) = 13}$$

$$g(8) = 8 + 5$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$\boxed{g^{-1}(0) = 1}$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g^{-1}(2) = 2 \quad g(2) = 2 \quad \boxed{g(g^{-1}(2)) = 2}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(1) = 6, \quad f(3) = 6$$

$$g(4) = 1, \quad g(0) = 3$$

$$\boxed{x = 0, x = 4 \text{ for}}$$

$$f(g(x)) = 6.$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{C - 597}{30}$$

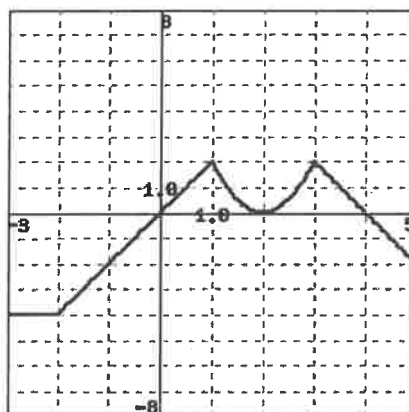
$$C = 30x + 597$$

$$- 597 \quad - 597$$

$$g^{-1}(C) = \frac{C - 597}{30}$$

$$\frac{C - 597}{30} = x$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1) \quad p(2) = m(1) \quad p(2) = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5 \quad w(2.5) = m(2) - 2.5 \quad w(2.5) = m(0.5) - 2.5$$

$$w(2.5) = 1$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) \quad k(1.5) = m(3) \quad k(1.5) = 2$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 \quad n(-2.3) = -0.3$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

Shift right by 2

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

up by 3

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

left 5, down 4

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) = (-3)^2 - 5 = 4$$

$$g(-3) = 4(-3) + 1 = -11$$

$$g(-3) = 4(-3) + 1 = -11$$

$$(f \cdot g)(-3) = -44$$

(b) (4 pts) $(f - g)(2)$

$$f(2) = (2)^2 - 5 = -1$$

$$g(2) = 4(2) + 1 = 9$$

$$g(2) = 4(2) + 1 = 9$$

$$(f - g)(2) = -10$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$f(0) = (0)^2 - 5 = -5$$

$$g(0) = 4(0) + 1 = 1$$

$$g(0) = 4(0) + 1 = 1$$

$$\left(\frac{f}{g}\right)(0) = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Jaslene Marquez WA3230981

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$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

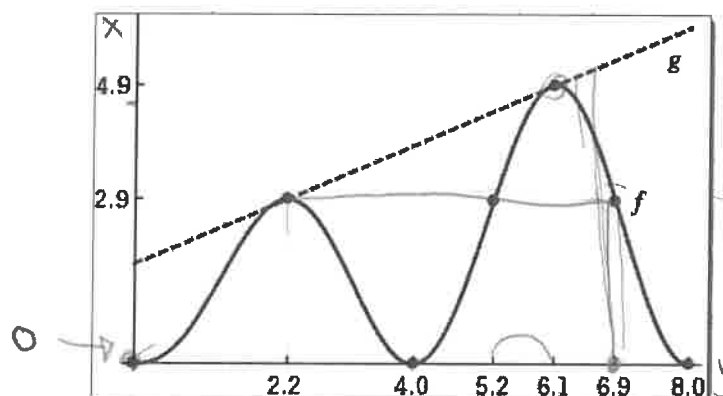
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 3.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 1.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 4.0, 8.0$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 2.2, 5.2, 6.9$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

The average rate of change of $f(x)$ is 5.2, 6.1 because it is ahead two numbers and it x. $4.0 \leq x \leq 6.9$

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

average rate
5.2, 6.1

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

17 inches long 5 inches wide

20 inches wide in total

$$V = 20 \text{ inches wide}$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$V(x) = (a, b) \sqrt{20 \text{ inches}}$$

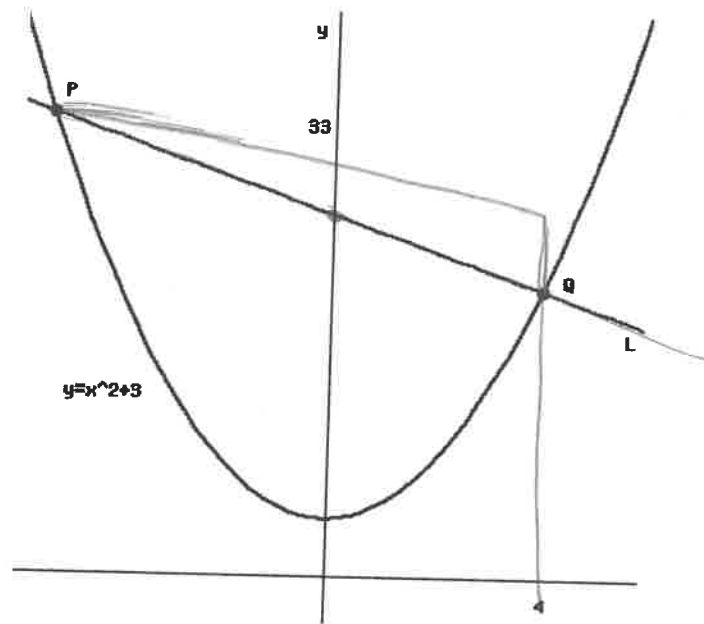
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

$$28, (1, 5)$$

$$x = 140$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



$$y = x^2 + 3$$

$$q = 4$$

$$p = 33$$

- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = 1089$$

$$Q = 19$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$Q = 10$$

$$P = 1089$$

$$L = 13$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = 10x^2 + 1089x^3 = L = 13$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$f(x) = x^2 + 5x - 14$$


This is the vertex

$$g(x) = a(x - h)^2 + k - 14$$

(13, 14)

- (b) (2 pts) Determine the y -intercept of the parabola.

y intercept is 14

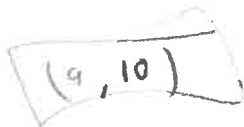
6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 3(x - 5)^2 - 9$$

$$g(x) = 3(x^2 - 5)^2 - 9$$


(9, 10)

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(x) = 2x - 7 \text{ and } g(x) = x + 5$$

$$(f) \quad x = 9, \quad (g)x = 7 \quad \underline{21}$$

(b) (3 pts) $(g \circ f)(3)$

$$g(1), f(9) \quad \underline{21}$$

(c) (3 pts) $(f \circ f)(3)$

$$f(9) = f(9) \quad \underline{27}$$

(d) (3 pts) $(g \circ g)(3)$

$$g(7)g(7) \quad \underline{16}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) \quad \underline{-1}$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2)) \quad \underline{-2}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$g(x) = 6$$

$$\boxed{3, 2 = 6}$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

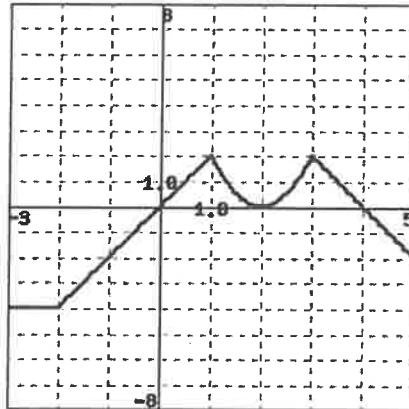
$$g^{-1}(C) =$$

$$g^{-1}(C)$$

$$g(x) = 30x + 597$$

$$x = 627$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = 1.25$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = 2.25$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = 2.3$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point negative number

$$x = -25$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point positive

$$x = 303$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point negative

$$x = 6.25$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(x) = x^2 - 5 \quad g(x) = 4x + 1$$

$$\rightarrow \sqrt{-3} \quad 62.6$$

(b) (4 pts) $(f - g)(2)$

$$(-5)(4)$$

$$3.25$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\frac{f}{g}(0) \quad \frac{4}{5}(0)$$

$$0$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Nunez Valdivia, Alexa WA 3284612

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

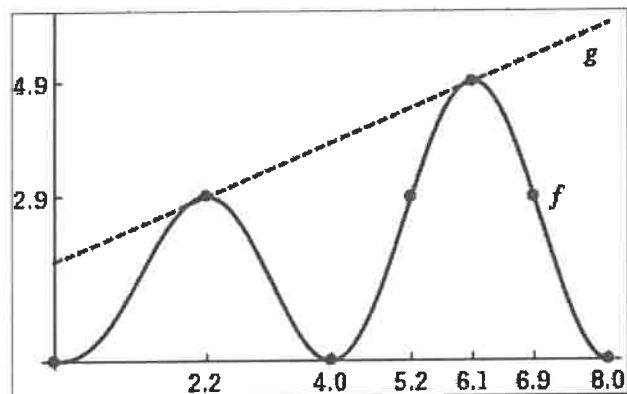
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4.0, 8.0$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

(0, 4.0) (2.9, 6.9) $AV = \frac{6.9 - 4.0}{2.9 - 0} = \frac{2.9}{2.9} = 1$

the average rate of change is 1.

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V(x) = x(17 - 2x)(5 - 2x)$$

$$L = 17 - 2x$$

$$W = 5 - 2x$$

$$H = x$$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$\text{domain } V(x) : (0, 2.5)$$

$$L = 17 - 2x \Rightarrow x < 8.5$$

$$W = 5 - 2x \Rightarrow x < 2.5$$

$$H = x > 0$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[1,5]} = \frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

$$f(1) = 36 - (1)^2$$

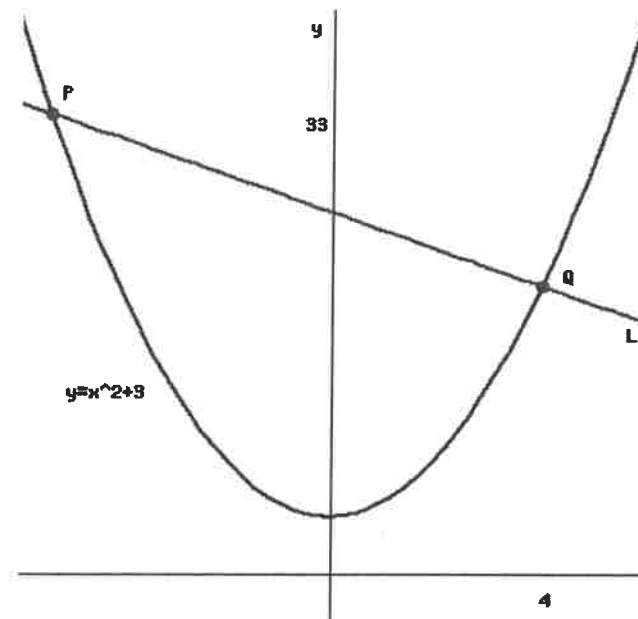
$$= 36 - 1 = 35$$

$$f(5) = 36 - (5)^2$$

$$= 36 - 25 = 11$$

$$AV_{[1,5]} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

$$\begin{aligned} 33 &= x^2 + 3 \\ -3 &\quad -3 \\ \hline \sqrt{30} &= \sqrt{x^2} \end{aligned} \quad \boxed{x = \sqrt{30}}$$

$$y = 4^2 + 3$$

$$y = 16 + 3$$

$$\boxed{y = 19}$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$m = \frac{19 - 33}{4 - (-\sqrt{30})} = \frac{-14}{4 - \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$\boxed{y = 19 + \frac{-14}{4 - \sqrt{30}}(x - 4)}$$

$$(4, 19)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$a = 1 \quad b = 5 \quad c = -14$$

$$x = \frac{-b}{2a} = \frac{-5}{2} = -2.5$$

$$\text{vertex: } (-2.5, -20.25)$$

$$\begin{aligned} f(-2.5) &= (-2.5)^2 + 5(-2.5) - 14 \\ &= 6.25 + (-12.5) - 14 \end{aligned}$$

$$y = -20.25$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$y\text{-intercept: } (0, -14)$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\begin{array}{ccc} 0 & = & 3(x-5)^2 - 9 \\ +9 & & +9 \end{array}$$

$$\frac{9}{3} = \frac{3(x-5)^2}{3}$$

$$\sqrt{3} = \sqrt{(x-5)^2}$$

$$x = \sqrt{3} + 5$$

$$x = -\sqrt{3} + 5$$

$$\begin{array}{ccc} \pm \sqrt{3} & = & x - 5 \\ +5 & & +5 \end{array}$$

$$\pm \sqrt{3} + 5 = x$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$g(3) = 3 + 5 = 8$$

$$(f \circ g)(3) = 9$$

$$f(8) = 2(8) - 7$$

$$= 16 - 7 = 9$$

(b) (3 pts) $(g \circ f)(3)$

$$f(3) = 2(3) - 7$$

$$= 6 - 7 = -1$$

$$(g \circ f)(3) = 4$$

$$g(-1) = -1 + 5 = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f(3) = 2(3) - 7$$

$$= 6 - 7$$

$$= -1$$

$$f(-1) = 2(-1) - 7$$

$$= -2 - 7$$

$$= -9$$

$$(f \circ f)(3) = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$g(3) = 3 + 5$$

$$= 8$$

$$g(8) = 8 + 5$$

$$= 13$$

$$(g \circ g)(3) = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g^{-1}(2) = 2$$

$$g(2) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$x = 0, 4$$

$$g(0) = 3$$

$$g(4) = 1$$

$$f(3) = 6$$

$$f(1) = 6$$

$$f(g(0)) = 6$$

$$f(g(4)) = 6$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{x - 597}{30}$$

$$g(x) = 30x + 597$$

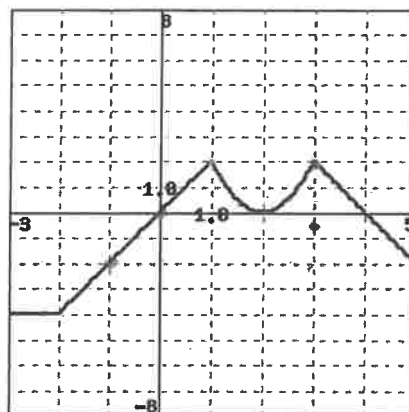
$$y = 30x + 597$$

$$x = \frac{y - 597}{30}$$

$$\frac{x - 597}{30} = \frac{y - 597}{30}$$

$$\frac{x - 597}{30} = y$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1) = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5 = -0.5$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) = 0$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = -2$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

shifts right by 2

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

moves up by 3

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

moves down by 4
left by 5

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$\boxed{f(-3) \cdot g(-3) = -44}$$

$$4(-11)$$

$$\begin{aligned} f(-3) &= (-3)^2 - 5 \\ &= 9 - 5 \\ &= 4 \end{aligned}$$

$$\begin{aligned} g(-3) &= 4(-3) + 1 \\ &= -12 + 1 \\ &= -11 \end{aligned}$$

(b) (4 pts) $(f - g)(2)$

$$\boxed{f(2) - g(2) = -10}$$

$$-1 - 9 = -10$$

$$\begin{aligned} f(2) &= (2)^2 - 5 \\ &= 4 - 5 \\ &= -1 \end{aligned}$$

$$\begin{aligned} g(2) &= 4(2) + 1 \\ &= 8 + 1 \\ &= 9 \end{aligned}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\boxed{\frac{f(0)}{g(0)} = \frac{-5}{1} = -5}$$

$$\begin{aligned} f(0) &= (0)^2 - 5 \\ &= 0 - 5 \\ &= -5 \end{aligned}$$

$$\begin{aligned} g(0) &= 4(0) + 1 \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Wilson Liu

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$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

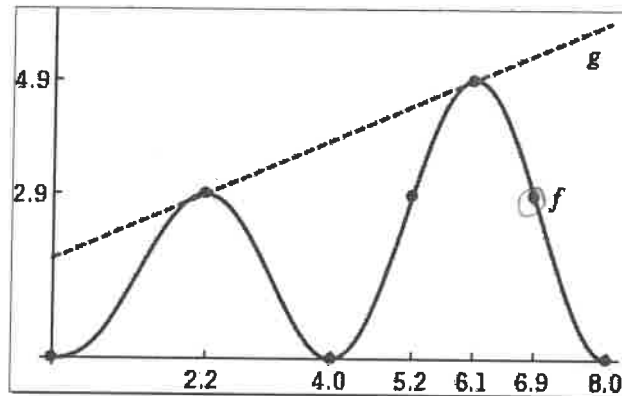
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4, 8$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$\frac{2.9 - 0}{6.9 - 4.0} = \frac{2.9}{2.9} = 1$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = (17 - 2x)(5 - 2x)$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$\begin{array}{lcl} 17 - 2x = 0 & & 5 - 2x = 0 \\ -17 = -2x & & -5 = -2x \\ \frac{-17}{-2} & & \frac{-5}{-2} \\ 8.5 = x & & 2.5 = x \end{array}$$

$$(2.5, 8.5)$$

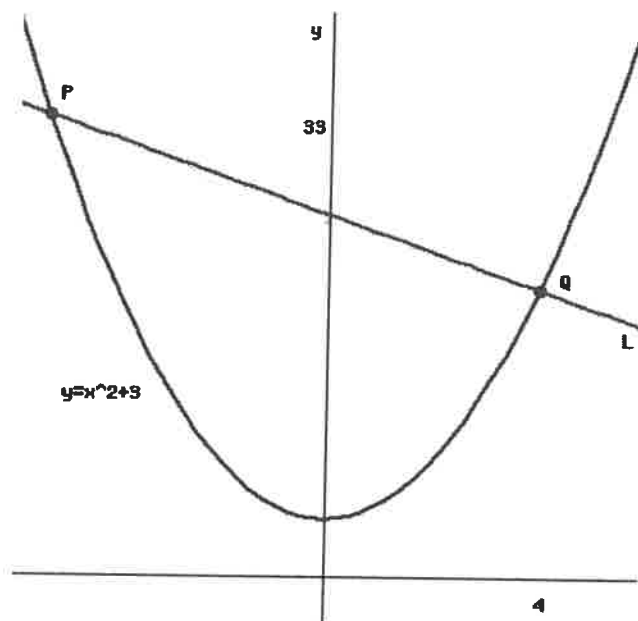
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(1) = 36 - 1^2 = 36 - 1 = 35$$

$$f(5) = 36 - 5^2 = 36 - 25 = 11$$

$$\frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (\sqrt{30}, 33)$$

$$Q = (4, 19)$$

$$4^2 + 3 = 19$$

$$33 = x^2 + 3$$

$$x = \sqrt{30}$$

$$30 = x^2$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\frac{19 - 33}{4 - \sqrt{30}} = -8.97$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$f(x) = -8.97(x - 4) + 19$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$V = \left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$$

$$\left(-\frac{5}{2(1)}, f\left(-\frac{5}{2(1)}\right)\right) = (-2.5, -2.5)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$y\text{-intercept} = c$$

$$c = -14$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$0 = 3(x - 5)^2 - 9$$

$$9 = 3(x - 5)^2$$

$$3 = (x - 5)^2$$

$$\sqrt{3} = x - 5$$

$$x = 5 + \sqrt{3}$$

$$x = 5 + \sqrt{3} = 6.7$$

$$x = 5 - \sqrt{3} = 3.26$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$3 + 5 = 8$$

$$f(x) = 2(8) - 7 = 9$$

(b) (3 pts) $(g \circ f)(3)$

$$2(3) - 7 = -1$$

$$g(x) = -1 + 5 = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$2(3) - 7 = -1$$

$$2(-1) - 7 = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$3 + 5 = 8$$

$$g(x) = 8 + 5 = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g^{-1}(2) = 1 \quad g(1) = 0$$

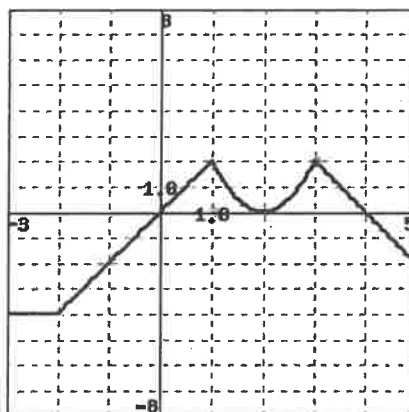
(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(g(x)) = 6$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = 19.9 - \frac{1}{30}C$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2-1) = m(1) = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5 = m(2) - 2.5 = 0 - 2.5 = -2.5$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) = m(3) = 2$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = -4 + 2 = -2$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$-3^2 - 5 =$$

$$9 - 5 = 4$$

$$4(-3) + 1 = 4(-11) = -44$$

$$-12 + 1 = -11$$

(b) (4 pts) $(f - g)(2)$

$$2^2 - 5 = 4 - 5 = -1$$

$$4(2) + 1 = 8 + 1 = 9$$

$$-1 - 9 = -10$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$0^2 - 5 = -5$$

$$4(0) + 1 = 1$$

$$\frac{-5}{1} = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: CODY-JACOB SANCHEZ #11432816

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

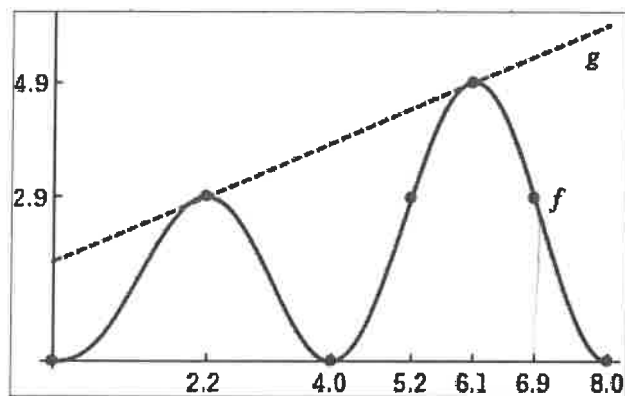
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = (0, 4.0, 8.0)$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = (2.2, 6.1)$$

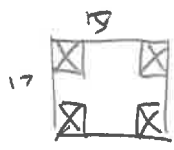
- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$\text{Avg}_{[a-b]} = \frac{f(b) - f(a)}{b - a} \Rightarrow \frac{f(6.9) - f(4.0)}{6.9 - 4.0} = \frac{2.9 - 0}{2.9}$$

$$= \boxed{2.9}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.



$$\begin{aligned} L &= 17 - x - x &> 17 - 2x \\ W &= 5 - x - x &> 5 - 2x \\ H &= x \end{aligned} \quad (l \cdot w \cdot h)$$

$$V(x) = (17 - 2x)(5 - 2x)x$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

*MUST BE POSITIVE

$$0 < x < 2.5 \text{ (strict)}$$

$$x > 0 \rightarrow h \Rightarrow$$

$$5 - 2x > 0 \rightarrow w \Rightarrow 5/2 = x < 2.5$$

$$17 - 2x > 0 \rightarrow L \Rightarrow 17/2 = x < 8.5$$

$(0, 2.5)$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

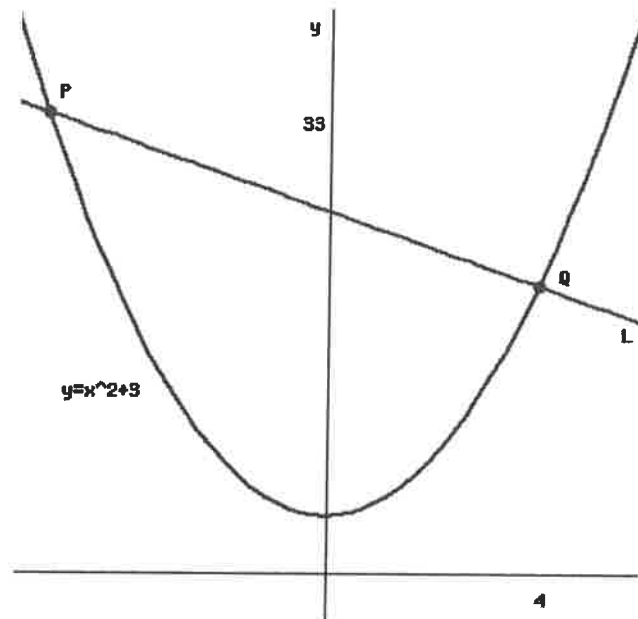
$$Av[a-b] = \frac{f(b) - f(a)}{b - a} = \frac{f(5) - f(1)}{5 - 1} = \frac{-24}{4} \text{ or } -6$$

$$f(1) = 36 - 1^2 = 35$$

$$f(5) = 36 - 5^2 = -4$$



4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = 33 = x^2 + 3 = 30 = x^2; \boxed{\pm \sqrt{30}} \quad P \sqrt{30}$$

$$Q = 4^2 + 3 = \boxed{19} \quad Q \sqrt{19}$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$M = \frac{19 - 33}{4 - \sqrt{30}} = \boxed{\frac{-14}{4 - \sqrt{30}}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$\text{POINT SLOPE} \rightarrow \boxed{y = 19 + (4 - \sqrt{30})(x - 4)}$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

Vertex is: $X = \frac{-b}{2a}$ where $a = 1, b = 5$

Thus, $\frac{-5}{2(1)} = \boxed{\frac{-5}{2}}$ or $\boxed{-2.5}$

- (b) (2 pts) Determine the y -intercept of the parabola.

*Set $x = 0$

$$f(0) = 0^2 + 5(0) - 14$$

$$\boxed{y = (0, -14)}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

*Set $y = 0$

$$0 = 3(x - 5)^2 - 9 \quad (+9)$$

$$9 = 3(x - 5)^2 \quad (\div 3)$$

$$3 = (x - 5)^2 \quad (\text{square roots})$$

$$\boxed{x = 5 \pm \sqrt{3}}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(g(3)) \dots \quad f(8) = 2(8) - 7 = \boxed{9}$$

$$g(3) = 3 + 5 = 8 \rightarrow$$

(b) (3 pts) $(g \circ f)(3)$

$$g(f(3)) \quad f(3) = 2(3) - 7 = -1 \rightarrow g(-1) = -1 + 5 = \boxed{4}$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3)) \quad f(-1) = 2(-1) - 7 = \boxed{-9}$$

$$f(3) = 2(3) - 7 = -1 \rightarrow$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3)) \quad g(8) = 8 + 5 = \boxed{13}$$

$$g(3) = 3 + 5 = 8 \rightarrow$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

*Where does $g(x) = 0$? $\rightarrow g(1) = 0$

$$\text{Thus, } g^{-1}(0) = \boxed{1}$$

(b) (2 pts) Find $g^{-1}(2)$

$$g^{-1}(2)? \rightarrow g(2) = 2 \quad \text{Thus, } g^{-1}(2) = 2 \quad \text{or } g(2) = \boxed{2}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

*WHAT MAKES $f(x) = 6$?

$$f(1) = 6 \quad f(3) = 6 \quad \text{Thus } g(x) = 1 \quad \text{or } 3$$

$$g(4) = 1 \quad g(0) = 4 \quad \text{Thus } x = (0, 4) \text{ or } (4, 0)$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

1.) write $y = f(x)$

2.) solve for 'x'

3.) rename $f^{-1}(x)$

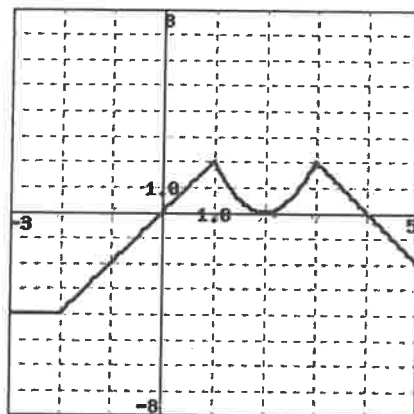
$$(1) \quad C = 30x + 597 \quad (\text{subtract})$$

$$C - 30x = 597$$

(2)

$$x = \frac{C - 597}{30} ; \quad g^{-1}(C) = \frac{C - 597}{30} \quad (3)$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2)$

$$p(t) = m(t - 1) \rightarrow p(2) = m(2 - 1) : m(1) = \boxed{2}$$

(b) (3 pts) $w(2.5)$

$$w(t) = m(t - 0.5) - 2.5 \rightarrow m(2.5 - 0.5) = m(2) = 0$$

$$0 - 2.5 = \boxed{-2.5}$$

(c) (3 pts) $k(1.5)$

$$k(t) = m(t + 1.5) \rightarrow k(1.5) = m(1.5 + 1.5) : m(3) = \boxed{2}$$

(d) (3 pts) $n(-2.3)$

$$n(t) = m(t) + 2 \rightarrow n(-2.3) = m(-2.3) + 2 : m(-2.3) =$$

$$-4 + 2 = \boxed{-2}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

inside: $(x - 2)$
 $-x$ shifts right by 2 $> (-5 + 2, 7)$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

outside: $+3$
 y shifts up by 3 $> (-5, 3 + 7)$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(0, 3)$

inside: $x + 5$
 outside: -4 $> (-5 - 5, 7 - 4)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f = (-3)^2 - 5 = 4$$

$$g = 4(-3) + 1 = -11$$

$$> 4(-11) = \boxed{-44}$$

(b) (4 pts) $(f - g)(2)$

$$f = (2)^2 - 5 = -1$$

$$g = 4(2) + 1 = 9$$

$$> -1 - 9 = \boxed{-10}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$f = (0)^2 - 5 = -5$$

$$g = 4(0) + 1 = 1$$

$$\frac{-5}{1} = \boxed{-5}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Jow, Anthony WA3277268

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
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- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

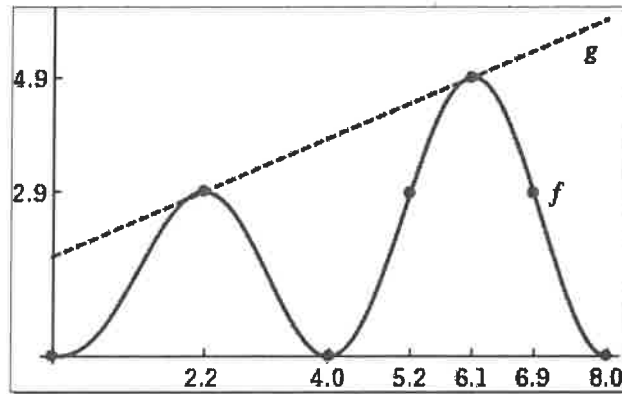
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9)

(6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4, 8$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

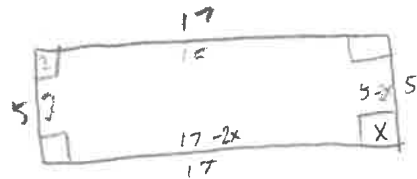
$$\frac{2.9 - 0}{6.9 - 4} = 1$$

$AV_{[4.0, 6.9]} = 1$ for $f(x)$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = x \cdot (17 - 2x) \cdot (5 - 2x)$$



(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$\text{Domain} = (0, 2.5)$$

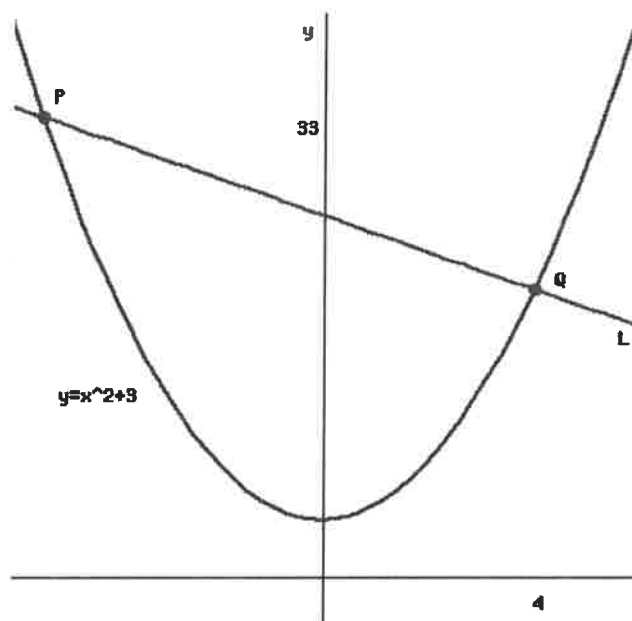
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$\frac{11 - 35}{5 - 1} = -\frac{24}{4}$$

$$= -6$$

$$AV_{[1,5]} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (\sqrt{30}, 33)$$

$$Q = (4, 19)$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\frac{\sqrt{30} - 4}{33 - 19} = \frac{\text{slope}}{\sqrt{30} - 4}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$L = 19 + \frac{\sqrt{30} - 4}{14} (x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14. \quad (x+2.5)^2 - 20.25$$

- (a) (6 pts) Determine the vertex of the parabola.

Vertex
 $(-2.5, -20.25)$

$$(0, -14)$$

$$(-1, -18)$$

$$(-2, -20)$$

$$(-3, -20)$$

$$(-2.5, -20.25)$$

$$-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)$$

$$-\frac{5}{2}, f\left(-\frac{5}{2}\right)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

y -intercept
 $(0, -14)$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x-5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$3(x-5)^2 - 9 = 0$$

$$x = \sqrt{3} + 5, -\sqrt{3} + 5$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$ $f(g(3)) = f(8)$

$$= 9$$

(b) (3 pts) $(g \circ f)(3)$

$$g(f(3)) = g(-1)$$

$$= 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3)) = f(-1)$$

$$= -10$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3)) = g(8)$$

$$= 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2)) = g(2)$$

$$= 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f(g(x)) = 6$$

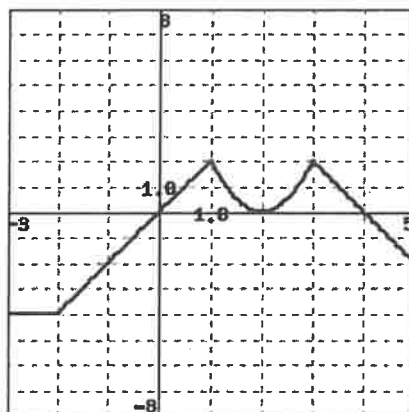
$$g(x) = 3, 1$$

$$x = 0, 4$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = -\frac{597}{30x}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1) = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5$$

- (c) (3 pts) $k(1.5)$

$$= -2.5$$

$$k(1.5) = m(1.5 + 1.5) = 2$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2$$

$$= -2$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

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(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) \cdot g(-3) = 4 \cdot -11 \\ = -44$$

(b) (4 pts) $(f - g)(2)$

$$f(2) - g(2) = -1 - 9 \\ = -10$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\frac{f(0)}{g(0)} = \frac{-5}{1} \\ = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Lwin Kyaw Vun WA 3265399

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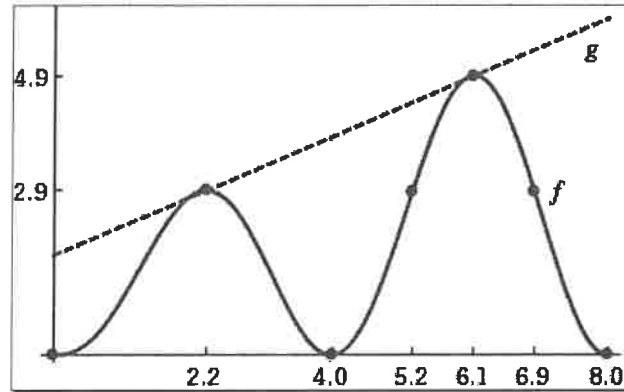
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$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right)\right)$$

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1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = 2.9$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$ $(6.9, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4, 8$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$V_{[4, 6.9]} = \frac{2.9 - 0}{6.9 - 4} = \frac{2.9}{2.9}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V(x) = x(17-2x)(5-2x)$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$x > 0 \quad 17 - 2x > 0 \quad 5 - 2x > 0$$

$$0 < x < 2.5$$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$f(x) = 36 - x^2$$

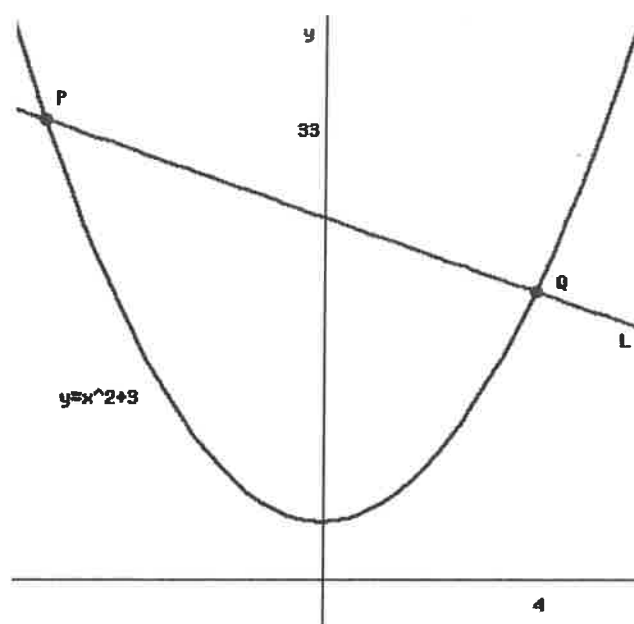
$$f(1) = 36 - 1 = 35$$

$$f(5) = 36 - 25 = 11$$

$$\frac{f(5) - f(1)}{5 - 1} = \frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

$$\frac{f(5) - f(1)}{5 - 1} = \frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

stop

at Q , $x = 4$

$$y = x^2 + 3 = 4^2 + 3$$

$$= 16 + 3 = 19$$

at P , $y = 33$

$$33 = x^2 + 3$$

$$x^2 = 30$$

$$x = \pm\sqrt{30}$$

$x = -\sqrt{30}$ on the left side of the y -axis

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\text{slope} = \frac{19 - 33}{4 - (-\sqrt{30})} = \frac{-14}{4 + \sqrt{30}} = \frac{-14(4 - \sqrt{30})}{(4 + \sqrt{30})(4 - \sqrt{30})} = \frac{-14(4 - \sqrt{30})}{16 - 30}$$

$$m = (4 - \sqrt{30})$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = 19 + (4 - \sqrt{30})(x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$\begin{aligned}x &= -\frac{b}{2a} = -\frac{5}{2} \\f\left(-\frac{5}{2}\right) &= \left(-\frac{5}{2}\right)^2 + 5\left(-\frac{5}{2}\right) - 14 \\&= \frac{25}{4} - \frac{25}{2} - 14 \\&= \frac{25 - 50 - 56}{4} = -\frac{81}{4} \\&\boxed{\text{vertex } \left(-\frac{5}{2}, -\frac{81}{4}\right)}\end{aligned}$$

- (b) (2 pts) Determine the y-intercept of the parabola.

$$\begin{aligned}\text{y intercept } x &= 0 \\y &= x^2 + 5x - 14 = -14 \\&\boxed{(0, -14)}\end{aligned}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\begin{aligned}3(x - 5)^2 - 9 &= 0 \\3(x - 5)^2 &= 9 \\(x - 5)^2 &= 3 \\x - 5 &= \pm\sqrt{3} \\x &= \pm\sqrt{3} + 5 \\&\boxed{x = \sqrt{3} + 5} \text{ or } \boxed{x = -\sqrt{3} + 5}\end{aligned}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(g(3)) = f(8) = 16 - 7 = 9$$

$$(f \circ g)(3) = 9$$

(b) (3 pts) $(g \circ f)(3)$

$$g(f(3)) = g(-1) = 4$$

$$(g \circ f)(3) = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3)) = f(-1) = -9$$

$$(f \circ f)(3) = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3)) = g(8) = 13$$

$$(g \circ g)(3) = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2)) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$x = 0, 4$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

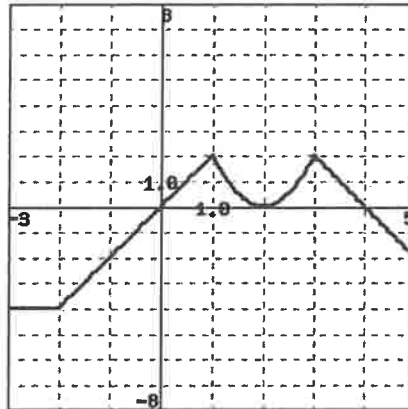
$$g^{-1}(C) = ?$$

$$C = 30x + 597$$

$$x = \frac{C - 597}{30}$$

$$g^{-1}(C) = \frac{C - 597}{30}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2) = m(2 - 1) = m(1) = 2$

(b) (3 pts) $w(2.5) = m(2.5 - 0.5) - 2.5 = m(2) - 2.5 = 0 - 2.5 = -2.5$

(c) (3 pts) $k(1.5) = m(1.5 + 1.5) = m(3) = 2$

(d) (3 pts) $n(-2.3) = m(-2.3) + 2 = -4 + 2 = -2$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

$$(-5 + 2, 7)$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

$$(-5, 7 + 3)$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

$$(-5 - 5, 7 - 4)$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) = f(-3) \cdot g(-3)$
 $= 4 \cdot (-12) = -44$

$$(f \cdot g)(-3) = -44$$

(b) (4 pts) $(f - g)(2) = f(2) - g(2)$
 $= -1 - 9$
 $= -8$

$$(f - g)(2) = -8$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0) = \frac{f(0)}{g(0)} = \frac{-5}{1} = -5$

$$\left(\frac{f}{g}\right)(0) = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Keys, Lio WA3179760

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and **box all final answers**.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

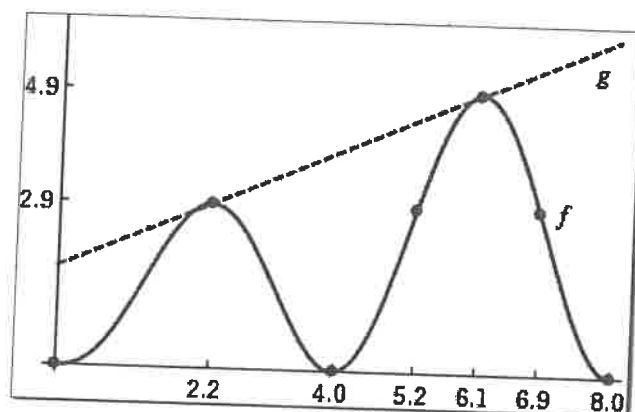
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



(a) (2 pts) Find $f(6.9) = \boxed{2.9}$

- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, $\boxed{2.9}$)

($\boxed{6.1}$, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = \boxed{0, 4.0, 8.0}$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = \boxed{2.2, 6.1}$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

AV_[4.0, 6.9] = $\frac{f(6.9) - f(4.0)}{6.9 - 4.0} = \frac{2.9 - 0}{2.9} = \frac{2.9}{2.9} = \boxed{1}$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V = f(x) = 17 \cdot 5 \cdot x$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$(5, 17)$$

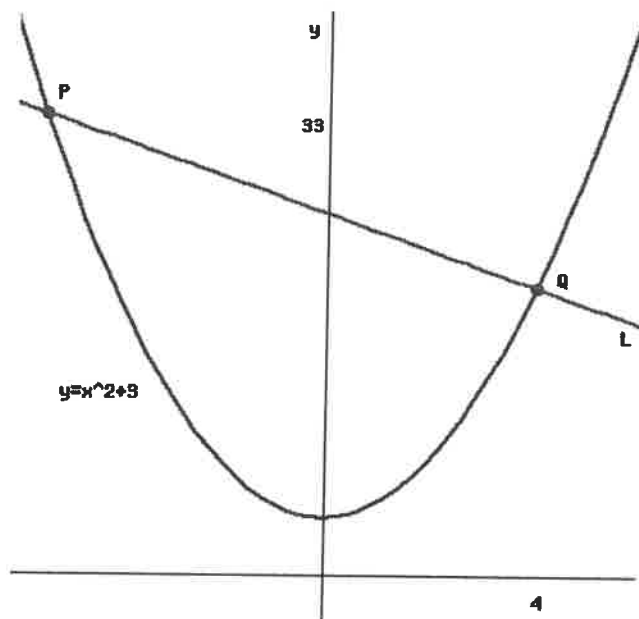
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[1,5]} = \frac{f(5) - f(1)}{5 - 1} = \frac{11 - 35}{4} = \frac{-24}{4} = -6$$

$$f(5) = 36 - (5)^2 \\ = 11$$

$$f(1) = 36 - (1)^2 \\ = 35$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-19, 33)$$

$$Q = (4, 54)$$

$$y = (-19)^2 + 3 \quad 33 = x^2 + 3$$

$$= 19 \quad 30 = x^2$$

$$\sqrt{30} = x$$

5.41

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\frac{54 - 33}{4 - (-19)} = \frac{27.6}{-15} = -1.84$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = -1.84x + 19$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right) \quad a=1, b=5, c=-14$$

$$\frac{-5}{2(1)} = \frac{-5}{2} = -2.5 \quad \rightarrow \quad f(-2.5) = (-2.5)^2 + 5(-2.5) - 14$$

$$= -20.25$$

$$\text{Vertex: } (-2.5, -20.25)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$f(x) = x^2 + 5x - 14$$

$$y = 0^2 + 5(0) - 14$$

$$= -14$$

$$y\text{-int: } (0, -14)$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$y = 3(x - 5)^2 - 9$$

$$0 = 3(x - 5)^2 - 9$$

$$9 = 3(x - 5)^2$$

$$3 = (x - 5)^2$$

$$\pm\sqrt{3} = x - 5$$

$$5 \pm \sqrt{3} = x$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$\begin{aligned} f(g(3)) &\rightarrow f(x) = 2(3+5) - 7 \\ &= 6 + 10 - 7 \\ &= 9 \end{aligned}$$

(b) (3 pts) $(g \circ f)(3)$

$$\begin{aligned} g(f(3)) &\rightarrow g(x) = (2(3) - 7) + 5 \\ &= 6 - 7 + 5 \\ &= 4 \end{aligned}$$

(c) (3 pts) $(f \circ f)(3)$

$$\begin{aligned} f(f(3)) &\rightarrow f(x) = 2(2(3) - 7) - 7 \\ &= 2(6 - 7) - 7 \\ &= -9 \end{aligned}$$

(d) (3 pts) $(g \circ g)(3)$

$$\begin{aligned} g(g(3)) &= g(x) = ((3) + 5) + 5 \\ &= 13 \end{aligned}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g(g^{-1}(2)) \rightarrow g^{-1}(2) = 2 \rightarrow g(2) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$\begin{aligned} f(3) &= 6 & f(1) &= 6 \\ g(0) &= 3 & g(4) &= 1 \end{aligned}$$

Values of x for $f(g(x)) = 6$ are 0, 4

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{x - 597}{30}$$

$$g(x) = 30x + 597$$

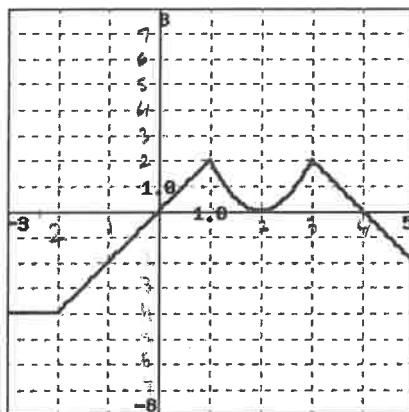
$$y = 30x + 597$$

$$x = \frac{y - 597}{30}$$

$$x - 597 = 30y$$

$$\frac{x - 597}{30} = y$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = m(2 - 1)$$

$$= m(1)$$

$$p(2) = 2$$

- (b) (3 pts) $w(2.5)$

$$w(2.5) = m(2.5 - 0.5) - 2.5$$

$$= m(2) - 2.5$$

$$= 2 - 2.5$$

$$= -0.5$$

$$w(2.5) = -2.5$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5)$$

$$= m(3)$$

$$k(1.5) = 2$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2$$

$$= 4 + 2$$

$$n(-2.3) = 6$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

$$-5 - 2 = -3$$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 4)$

$$7 - 3 = 4$$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 11)$

$$-5 - 5 = -10, 7 + 4 = 11$$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3) =$ -44

$$f(-3) \cdot g(-3)$$

$$f(-3) = (-3)^2 - 5 = 4$$

$$g(-3) = 4(-3) + 1 = -11$$

(b) (4 pts) $(f - g)(2) =$ -10

$$f(2) - g(2)$$

$$f(2) = (2)^2 - 5 = -1$$

$$g(2) = 4(2) + 1 = 9$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0) =$ -5

$$\frac{f(0)}{g(0)}$$

$$f(0) = (0)^2 - 5 = -5$$

$$g(0) = 4(0) + 1 = 1$$

$$\frac{-5}{1}$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Urauhart, Anne WA3267590

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Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

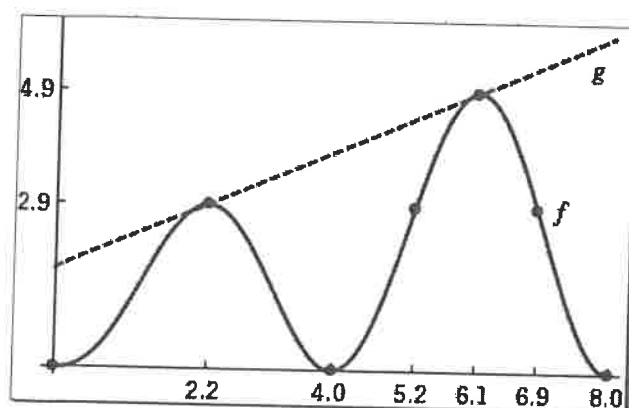
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = \boxed{2.9}$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$(2.2, 2.9)$

$(6.1, 4.9)$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 0, 4, 8$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x = 2.2, 6.1$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$? $\left(\frac{4.9}{2.9}\right) \left(\frac{2.9}{2.9}\right)$

$$AV_{[4, 6.9]} = \frac{2.9 - 0}{6.9 - 4} = \frac{2.9}{2.9} = \boxed{1}$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

$$V(x) = (17 - 2x)(5 - 2x)(x)$$

$$V = l \cdot w \cdot h \quad l = 17 - 2x \quad w = 5 - 2x \quad h = x$$

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

$$(0, 2.5)$$

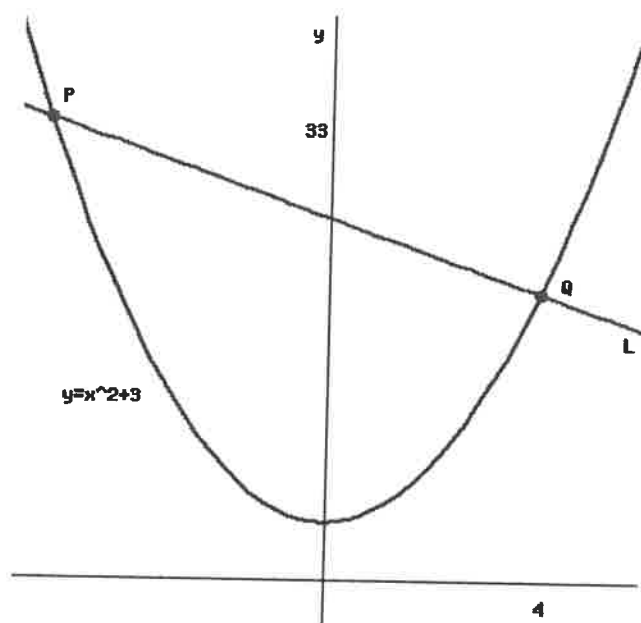
3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$AV_{[1,5]} = \frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

$$f(1) = 36 - 1^2 = 35 \quad (1, 35)$$

$$f(5) = 36 - 5^2 = 11 \quad (5, 11)$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33)$$

$$Q = (4, 19)$$

$$P: 33 = x^2 + 3$$

$$-\sqrt{30} = x$$

$$(-\sqrt{30}, 33)$$

$$Q: y = 4^2 + 3$$

$$y = 19$$

$$(4, 19)$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$AV_{[-\sqrt{30}, 4]} = \frac{19 - 33}{4 + \sqrt{30}} = \frac{-14}{4 + \sqrt{30}}$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = y_0 + m(x - x_0)$$

$$y = 19 + \left(\frac{-14}{4 + \sqrt{30}}\right)(x - 4)$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$\text{vertex: } (-2.5, -20.25)$$

$$-\frac{b}{2a} \quad a=1 \quad b=5 \quad c=-14$$

$$-\frac{5}{2 \cdot 1} = -2.5$$

$$\begin{aligned} f(-2.5) &= -2.5^2 + 5(-2.5) - 14 \\ &= 6.25 - 12.5 - 14 \\ &= -20.25 \end{aligned}$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$y\text{-int: } -14$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x-5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\text{zeros: } \left(\frac{30 - \sqrt{108}}{6}, \frac{30 + \sqrt{108}}{6} \right)$$

$$y = 3(x-5)^2 - 9 \Rightarrow 3(x^2 - 10x + 25) - 9 \Rightarrow 3x^2 - 30x + 66$$

$$a=3 \quad b=-30 \quad c=66$$

$$x = \frac{30 \pm \sqrt{-30^2 - 4(3)(66)}}{2 \cdot 3} \Rightarrow \frac{30 \pm \sqrt{108}}{6}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$g(3) = 8$$

$$f(g(3)) = 2(8) - 7 = \boxed{9}$$

(b) (3 pts) $(g \circ f)(3)$

$$f(3) = -1$$

$$g(f(3)) = -1 + 5 = \boxed{4}$$

(c) (3 pts) $(f \circ f)(3)$

$$f(f(3)) = 2(-1) - 7 = \boxed{-9}$$

(d) (3 pts) $(g \circ g)(3)$

$$g(g(3)) = 8 + 5 = \boxed{13}$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$\boxed{1}$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$\boxed{2}$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$\boxed{0 \text{ and } 4}$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{C}{30} - 19.9$$

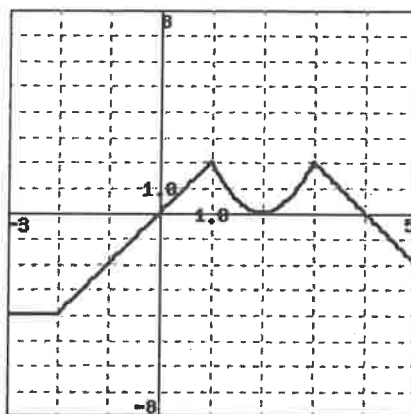
$$C = 30x + 597$$

$$C - 597 = 30x$$

$$\frac{C - 597}{30} = x$$

$$\frac{C}{30} - 19.9 = x$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

(a) (3 pts) $p(2) = \boxed{2}$

$$\begin{aligned} p(2) &= m(2-1) \\ &= m(1) \\ &= 2 \end{aligned}$$

(b) (3 pts) $w(2.5) = \boxed{0}$

$$\begin{aligned} w(2.5) &= m(2.5 - 0.5) \\ &= m(2) \\ &= 0 \end{aligned}$$

(c) (3 pts) $k(1.5) = \boxed{2}$

$$\begin{aligned} k(1.5) &= m(1.5 + 1.5) \\ &= m(3) \\ &= 2 \end{aligned}$$

(d) (3 pts) $n(-2.3) = \boxed{-2}$

$$\begin{aligned} n(-2.3) &= m(-2.3) + 2 \\ &= -4 + 2 \\ &= -2 \end{aligned}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(-3, 7)$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(-5, 10)$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(-10, 3)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$f(-3) = 4$$

$$g(-3) = -11$$

$$(f \cdot g)(-3) = -44$$

(b) (4 pts) $(f - g)(2)$

$$f(2) = -1$$

$$g(2) = 9$$

$$(f - g)(2) = -10$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$f(0) = -5$$

$$g(0) = 1$$

$$\left(\frac{f}{g}\right)(0) = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: Li, Justin, S00021792

Please read the following instructions carefully.

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Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

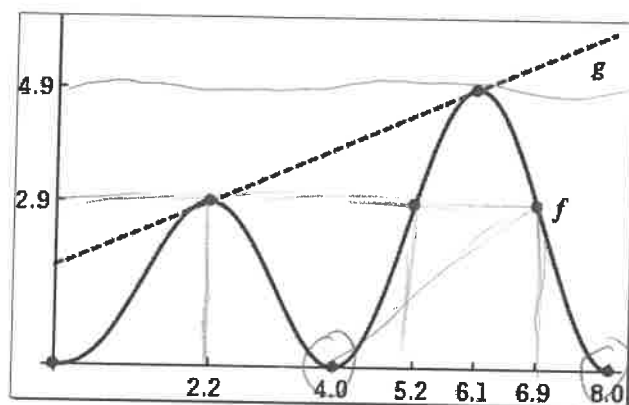
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

x y

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 4, 8$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = \text{None}$$

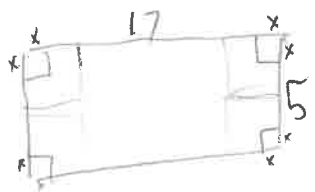
- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$? a b

$$\frac{2.9 - 0}{6.9 - 4} = \frac{2.9}{2.9} = 1$$

The average rate of change = 1

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.



$$A = 17 \cdot 5 = 85$$

$$4x^2$$

maybe its $x \cdot \frac{A}{85} - 4x^2$

- (b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

domain is $0 < x < 2.5$

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$36 - 1 = 35$$

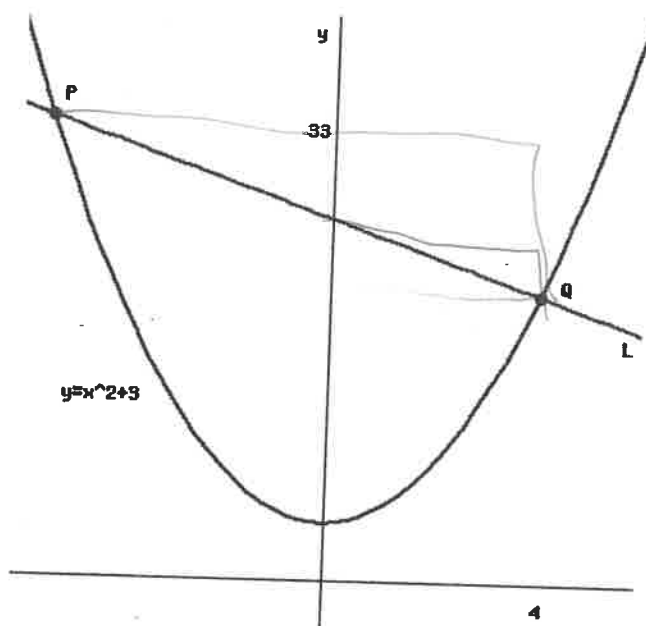
$$36 - 25 = 11$$



$$\frac{11 - 35}{5 - 1} = \frac{-24}{4} = -6$$

$$AV_{[1,5]} = -6$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = (-\sqrt{30}, 33) \quad P = 33 = x^2 + 3 \quad - (2\sqrt{30}) = -(\sqrt{30})$$

$$Q = (4, 19) \quad -3 = 30 \rightarrow \text{reverse pendos}$$

$$Q = 4^2 + 3 = ? = 16 + 3 = 19$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$\frac{f(Q) - f(P)}{Q - P} = \frac{19 - 33}{4 - (-\sqrt{30})} = \frac{-14}{4 + \sqrt{30}} \quad ?$$

slope?

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$f(L) = y = y_0 + m(x - x_0)$$

↓

$$19 = 4 + m(\sqrt{30} - 33) \quad ?$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14$$

- (a) (6 pts) Determine the vertex of the parabola.

5 right 14 down

$$+5x = 5 \text{ left}$$

$$-14 = 14 \text{ down}$$

The vertex is $(-5, -14)$ and $(0, -14)$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$(0, -14)$$

$$0^2 + 5 \cdot 0 - 14 = -14 \text{ or this}$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} =$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$f(g) = g = 3 + 5 = 8 \quad (f \circ g)(3) = 9$$

$$f(8) = 2 \cdot 8 - 7 = 9$$

(b) (3 pts) $(g \circ f)(3)$

$$f(3) = 6 - 7 = -1$$

$$g(-1) = -1 + 5 = 4 \quad (g \circ f)(3) = 4$$

(c) (3 pts) $(f \circ f)(3)$

$$f^1 = -1$$

$$f^2(-1) = -2 - 7 = -9 \quad (f \circ f)(3) = -9$$

(d) (3 pts) $(g \circ g)(3)$

$$g^2 = 8$$

$$g^2(8) = 8 + 5 = 13 \quad (g \circ g)(3) = 13$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

$$g^{-1}(0) = 1$$

(b) (2 pts) Find $g(g^{-1}(2))$

$$g^{-1}(2) = 2 \quad g(2) = 2 \quad g(g^{-1}(2)) = 2$$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

$$f^{-1}(6) = 3$$

$$g^{-1}(3) = 0$$

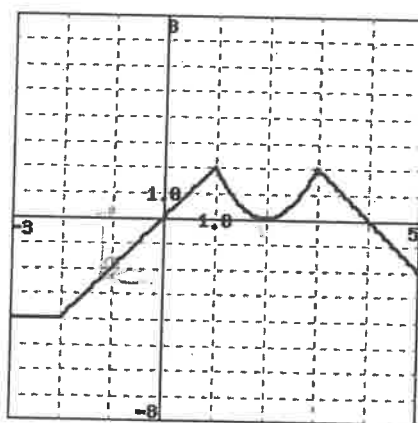
$$f(g(0)) = 3$$

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) = \frac{X - 597}{30}$$

$$\begin{aligned} g^{-1}(C) &= -X = 30 + 597 \\ &\quad - 597 \\ &\quad \downarrow \\ &\quad Y \\ X - 597 &= 30 \\ &\quad \downarrow \\ X - 597 & \\ \hline &30 \end{aligned}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$\begin{aligned} p(2) &= m(2 - 1) = m(1) = -2 \\ p(2) &= -2 \end{aligned}$$

- (b) (3 pts) $w(2.5)$

$$\begin{aligned} w(2.5) &= m(2.5 - 0.5) - 2.5 = m(2) - 2.5 = 0 - 2.5 = -2.5 \\ w(2.5) &= -2.5 \end{aligned}$$

- (c) (3 pts) $k(1.5)$

$$k(1.5) = m(1.5 + 1.5) = m(3) = 2$$

- (d) (3 pts) $n(-2.3)$

$$n(-2.3) = m(-2.3) + 2 = -2.5 + 2 = -0.5$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $-7, 7$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $-5, 10$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $0, 3$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$-3^2 - 5 \times -12 + 1 = -14 \times -11 = 154$$

(b) (4 pts) $(f - g)(2)$

$$\begin{matrix} (4, 2) \\ 4 - 5 - 8 + 1 = -1 - 8 = -10 \end{matrix}$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\frac{-5}{1} = -5$$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: De Haro , Alex WA

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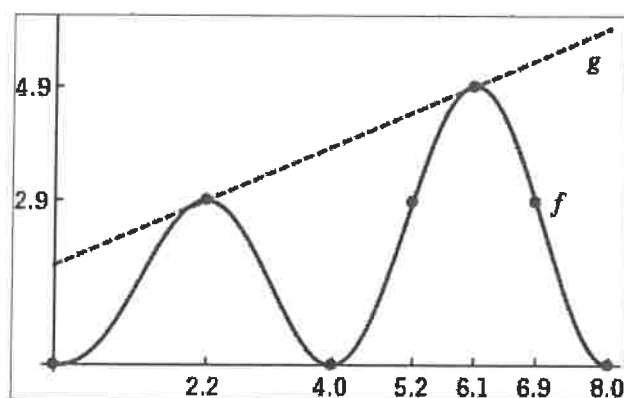
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
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1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(2.2, 2.9) (6.1, 4.9)

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$36 - 1^2 = 35$$

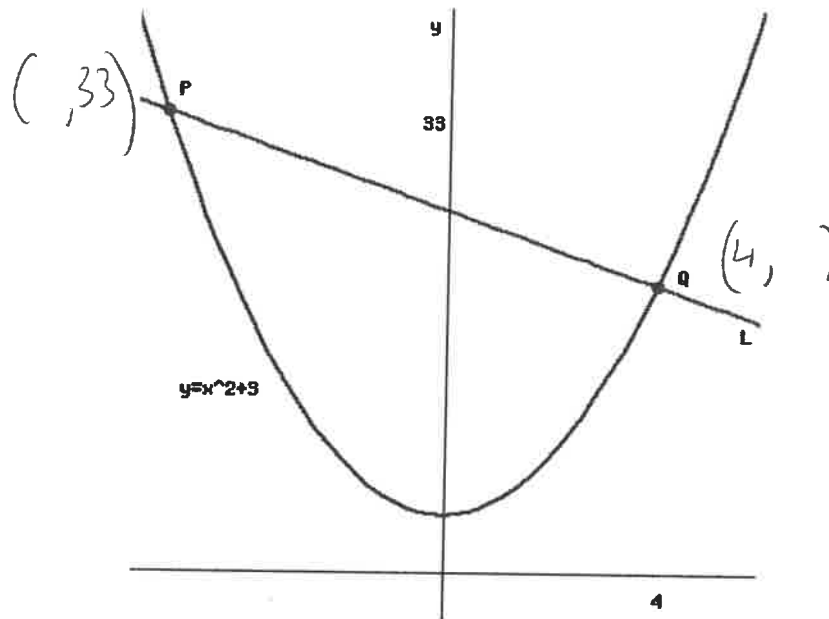
$$(1, 35)$$

$$36 - 5^2 = 11$$

$$(5, 11)$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{11 - 35}{5 - 1} = \frac{-24}{4} = \boxed{-6}$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$P =$

$Q =$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

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5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$\left(-\frac{b}{2a}, q\right)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$(2(3) - 7)(3 + 5) = (6 - 7)(3 + 5) = (-1)(8) = \boxed{-8}$$

(b) (3 pts) $(g \circ f)(3)$

$$(3 + 5)(2(3) - 7) = (8)(-1) = \boxed{-8}$$

(c) (3 pts) $(f \circ f)(3)$

(d) (3 pts) $(g \circ g)(3)$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

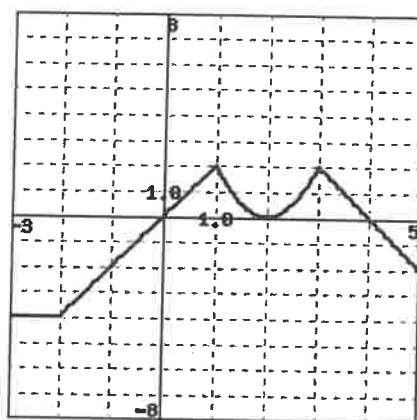
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$$g^{-1}(C) =$$

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Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

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(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point _____

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

(b) (4 pts) $(f - g)(2)$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

Chapter 1 Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#:

WA3242701 Brambila Kristin

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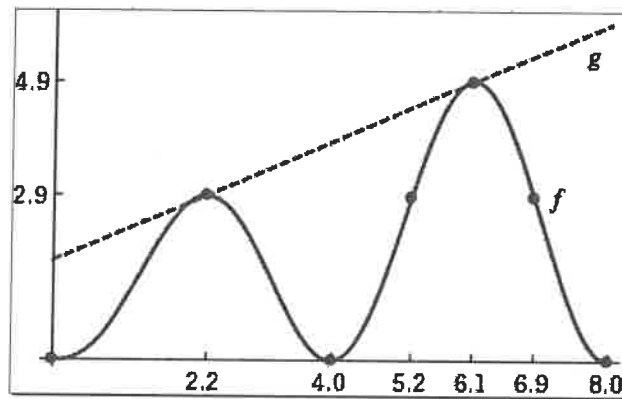
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

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1. (10 pts) The image shows the graphs of $f(x)$ and $g(x)$.



- (a) (2 pts) Find $f(6.9) = 2.9$
- (b) (2 pts) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

$$(2.2, 2.9) \quad (6.1, 4.9)$$

- (c) (2 pts) For what values of x is/are $f(x) = 0$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 8$$

- (d) (2 pts) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$$x = 2.9, 4.9$$

- (e) (2 pts) What is the average rate of change of $f(x)$ on the interval $4.0 \leq x \leq 6.9$?

$$\frac{6.9 - 4.0}{6.9 - 4.0} = \frac{2.9}{2.9} = 1$$

2. (8 pts) An open box is to be made from a flat piece of material 17 inches long and 5 inches wide by cutting equal squares of length x from the corners and folding up the sides.

(a) (6 pts) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

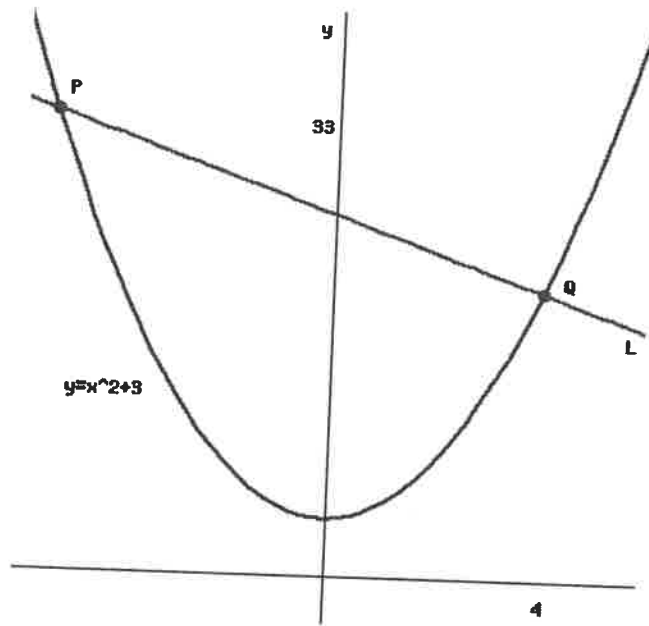
85

(b) (2 pts) Write the domain of the volume function V as an open interval in the form (a, b) .

3. (4 pts) Let $f(x) = 36 - x^2$. Compute the average rate of change on the interval $[1, 5]$. Write your answer as a positive or negative whole number.

$$\frac{5-1}{5-1} = \frac{4}{4} = 1$$

4. (10 pts) Consider the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola has equation $y = x^2 + 3$. Answer the following questions. Leave any radicals in radical form; do not approximate.



- (a) (4 pts) Determine the coordinates of the points P and Q .

$$P = 33$$

$$Q = 4$$

- (b) (4 pts) Use P and Q to determine the slope of the line L .

$$8.25$$

- (c) (2 pts) Use your answers from (a) and (b) to write an equation for L .

$$y = 8.25x^2 + 3$$

5. (8 pts) Consider the following quadratic function, whose graph is a parabola:

$$f(x) = x^2 + 5x - 14.$$

- (a) (6 pts) Determine the vertex of the parabola.

$$(5, -14)$$

- (b) (2 pts) Determine the y -intercept of the parabola.

$$-14$$

6. (6 pts) Find the zeros (if any) of the quadratic function

$$y = 3(x - 5)^2 - 9.$$

Leave any radicals in radical form; do not approximate.

$$\begin{aligned} y &= 3x - 15^2 - 9 \\ &= 3x + 225 - 9 \\ &= 3x + 216 \end{aligned}$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{+5 \pm \sqrt{-5^2 - 4(3)(-9)}}{2(3)}$$

$$\frac{+5 \pm \sqrt{25 - 12(-9)}}{6}$$

$$\frac{+5 \pm \sqrt{25 + 108}}{6}$$

$$\frac{+5 \pm \sqrt{133}}{6}$$

7. (12 pts) Given that $f(x) = 2x - 7$ and $g(x) = x + 5$, calculate the following.

(a) (3 pts) $(f \circ g)(3)$

$$(2x-7)(x+5)(3) \quad 2x^2 - 6$$

$$2x^2 - 2(3)$$

(b) (3 pts) $(g \circ f)(3)$

$$(x+5)(2x-7)(3) \quad 2x^2 - 6$$

$$2x^2 - 2(3)$$

(c) (3 pts) $(f \circ f)(3)$

$$(2x-7)(2x-7)(3) \quad 4x^2 - 42$$

$$4x^2 - 14(3)$$

(d) (3 pts) $(g \circ g)(3)$

$$(x+5)(x+5)(3) \quad 2x^2 + 30$$

$$2x^2 + 10(3)$$

8. (8 pts) Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	4	6	3	6	1
$g(x)$	3	0	2	4	1

(a) (2 pts) Find $g^{-1}(0)$

(b) (2 pts) Find $g(g^{-1}(2))$

(c) (4 pts) For what value(s) of x is $f(g(x)) = 6$?

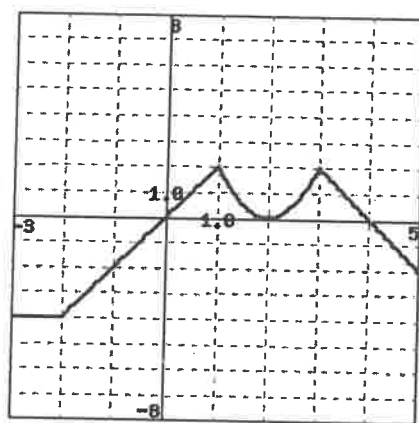
4

9. (4 pts) The cost (in dollars) of producing x air conditioners is $C = g(x) = 30x + 597$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

$$\begin{aligned} g(-1) &= 30(-1) + 597 \\ &= -30 + 597 \\ &= 567 \end{aligned}$$

10. (12 pts) The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) (3 pts) $p(2)$

$$p(2) = 2(2-1) = 2(1) = 2$$

- (b) (3 pts) $w(2.5)$

$$\begin{aligned} w(2.5) &= 2.5(2.5 - 0.5) \\ &= 2.5(2) = 5 \end{aligned}$$

- (c) (3 pts) $k(1.5)$

$$\begin{aligned} 1.5 &= 1.5(1.5 + 1.5) \\ 1.5(3) &= 4.5 \end{aligned}$$

- (d) (3 pts) $n(-2.3)$

$$\begin{aligned} -2.3 &= -2.3(-2.3 + 2) \\ &= -2.3(-0.3) \\ &= 0.69 \end{aligned}$$

11. (6 pts) The graph of $f(x)$ contains the point $(-5, 7)$. What point must be on each of the following transformed graphs?

(a) (2 pts) The graph of $f(x - 2)$ must contain the point $(0, -2)$

(b) (2 pts) The graph of $f(x) + 3$ must contain the point $(0, 3)$

(c) (2 pts) The graph of $f(x + 5) - 4$ must contain the point $(5, -4)$

12. (12 pts) For $f(x) = x^2 - 5$ and $g(x) = 4x + 1$, find:

(a) (4 pts) $(f \cdot g)(-3)$

$$(x^2 - 5)(4x + 1)(-3) \quad 4x^3 + 15$$

$$4x^3 - 5(-3)$$

(b) (4 pts) $(f - g)(2)$

$$(x^2 - 5) - (4x + 1)$$

$$x^2 - 5 - 4x - 1$$

$$-3x^2 - 6$$

(c) (4 pts) $\left(\frac{f}{g}\right)(0)$

$$\frac{x^2 - 5}{4x + 1} = 3x^2 - 4(0)$$

$$3x^2$$